

Mathematics Stage 5 Core

Index Laws

Book 1 Extend and apply the index laws to variables
Simplify algebraic products and quotients using
index laws

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Syllabus Content

MA5-IND-C-01 simplifies algebraic expressions involving positive-integer and zero indices, and establishes the meaning of negative indices for numerical bases

Extend and apply the index laws to variables, using positive-integer indices and the zero index

- Apply the index laws for numerical bases with positive-integer indices to develop the index laws in algebraic form
- Establish that $x^0 = 1$ algebraically using index laws
- Simplify algebraic expressions that involve the zero index

Simplify algebraic products and quotients using index laws

- Simplify algebraic expressions that involve powers, products and quotients of simple algebraic terms containing positive-integer indices

Index Law of Multiplication

Review



Recall index notation

Write, showing repeated multiplication

a a^5

b 5^2

c m^4

d x^5



Recall index law of multiplication for numerical bases

a $2^3 \times 2$

b $2^3 \times 2^2$

c $7^3 \times 7^{12}$

d $2^3 \times 5^{12}$

e $x + x + x$

f $x \times x \times x$

Identify when to use index law of multiplication



$5^3 \times 7^2$



$5^3 \times 5^2$



$x^4 \times x^9$



$m \times m^3$



$m + m^3$

Identify whether you can apply the index law of multiplication

a $4^2 \times 7^3$

b $a^5 \times a^2$

c $x^5 \times y^2$

d $p \times p^6$

e $2^4 \times 3^4$

f $b^{10} \times b$

g $b^{10} + b$

h $x^5 \times y^5$

① Index Law of Multiplication

$$a^m \times a^n = a^{m+n}$$

② Simplify using index law of multiplication

$$x^3 \times x^4$$
$$= x^7$$

$$x^3 \times x^{40}$$
$$= x^{43}$$

$$y^3 \times y^{40}$$
$$= y^{43}$$

Simplify the following expressions

a $x^2 \times x^5$

b $a^6 \times a^3$

c $m \times m^7$

d $b^4 \times b^2 \times b^3$

e $x^2 \times y^3$

f $2^5 \times 2^3$

g $(x + 3)^7 \times (x + 3)^4$

h $y \times y$

i $y + y$

💡 Key Ideas

1. When multiplying powers with the same base, the powers.
2. $x^m \times x^n = \dots\dots\dots$
3. The law only applies when the base is the in both terms.

1 Find the missing numbers.

a $3^2 \times 3^4 = 3 \times \square \times 3 \times \square \times \square \times \square$
 $= 3^{\square}$

3^6

b $6^4 \times 6^3 = 6 \times \square \times \square \times \square \times 6 \times \square \times \square$
 $= 6^{\square}$

6^7

c $k^3 \times k^2 = k \times \square \times \square \times \square \times \square$
 $= k^{\square}$

k^5

d $m^4 \times m^3 = m \times \square \times \square \times \square \times \square \times \square \times \square$
 $= m^{\square}$

m^7

e $5^3 \times 5 = \square \times \square \times \square \times \square$
 $= \square^{\square}$

5^4

f $2 \times 2^4 = \square \times \square \times \square \times \square \times \square$
 $= \square^{\square}$

2^5

2 Explain why you can't simplify $x^2 \times y^3$ using index laws.

The bases are not the same.

3 For each, write $\checkmark$ if the index law of multiplication applies, or $\times$ if it does not.

a $5^3 \times 5^7$

$\checkmark$

b $m^5 \times m^2$

$\checkmark$

c $6^3 \times 6$

$\checkmark$

d $4^5 \times 4^2$

$\checkmark$

e $x^4 \times y^3$

$\times$

f $b \times b^9$

$\checkmark$

g $c \times d^2$

$\times$

h $h^3 \times h$

$\checkmark$

i $a^2 \times a^5$

$\checkmark$

j $x^2 \times y^2$

$\times$

k $n^5 \times n^3 \times n^2$

$\checkmark$

l $e^2 \times f^2$

$\times$

m $2^3 \times 3^4$

$\times$

n $p^4 \times p^7$

$\checkmark$

o $q^2 \times r^3$

$\times$

p $t^4 \times t \times t^3$

$\checkmark$

4 For each, write each as a single power.

a $x^2 \times x^3$

x^5

e $x^3 \times x^4$

x^7

i $x^5 \times x^2$

x^7

b $x^2 \times x^4$

x^6

f $x^3 \times x^5$

x^8

j $x^5 \times x^3$

x^8

c $x^2 \times x^5$

x^7

g $x^4 \times x^3$

x^7

k $x^6 \times x^2$

x^8

d $x^2 \times x^6$

x^8

h $x^4 \times x^5$

x^9

l $x^7 \times x^3$

x^{10}

5 For each, write each as a single power.

a $x \times x^2$

x^3

b $y^5 \times y$

y^6

c $b^2 \times b$

b^3

d $n^3 \times n \times n^2$

n^6

6 Write each of these using a single power.

a $x^2 \times x^5$	b $y^3 \times y^4$	c $a^5 \times a^6$
x^7	y^7	a^{11}
d $4^3 \times 4^5$	e $5^3 \times 5$	f $b^3 \times b^2 \times b^4$
4^8	5^4	b^9
g $e^3 \times e^4 \times e^6$	h $b^4 \times b \times b^2$	i $n^2 \times n^5 \times n^2 \times n^3$
e^{13}	b^7	n^{12}
j $x^4 \times x^{18}$	k $p^{35} \times p^{42}$	
x^{22}	p^{77}	

7 Laura is simplifying $e^4 \times e^3$.



You multiply the powers, so the answer is e^{12}

Explain why she is incorrect and state the correct answer.

When multiplying the same base numbers, we add the powers.

e^7

8 Julie is simplifying $w^4 \times w$



The answer is w^4 since the second w has no power.

Explain why she is incorrect and state the correct answer.

The second power has a power of 1.

w^5

9 Simplify, if possible.

a $w^5 \times w^2 \times w^3 \times w$	b $h^4 \times h^4 \times h^4$	c $p^{99} \times p$
w^{11}	h^{12}	p^{100}
d $q^{100} \times q^{100}$	e $s^{200} \times s^{300}$	f $m \times m^3$
q^{200}	s^{500}	m^4
g $h^2 + h^2$	h $m + m^3$	i $a^4 \times b^5$
$2h^2$	Can't simplify	$a^4 b^5$

10 Write as a single power, if possible.

a $7^9 \times 7^6$

b $y^9 \times y^6$

c $a^9 \times b^6$

7^{15}

y^{15}

Can't simplify

d $(2x + 3)^9 \times (2x + 3)^6$

e $(\sqrt{7})^9 \times (\sqrt{7})^6$

f $9^7 \times 6^7$

$(2x + 3)^{15}$

$(\sqrt{7})^{15}$

Can't simplify

11 Work out the missing powers.

a $x^3 \times x^{\boxed{}} = x^5$

2

b $4^{\boxed{}} \times 4^2 = 4^6$

4

c $y^5 \times y^{\boxed{}} = y^{17}$

12

d $h^{\boxed{}} \times h^4 = h^5$

1

e $3^4 \times 3^{\boxed{}} = 3^9$

5

f $k^{\boxed{}} \times k^5 = k^{19}$

14

g $10^4 \times 10^{\boxed{}} \times 10 = 10^8$

3

12 Write each of these expressions as a single term.

a $2^x \times 2^{3x}$

2^{4x}

b $5^{3x-1} \times 5^{7x}$

5^{10x-1}

c $r^{2y+3} \times r^{y-2}$

r^{3y+1}

13 Every answer is **incorrect**, state the error and write the correct answer.

a $x^3 \times x^4 = x^{12}$

Multiplied instead of added the exponents. Correct: x^7

b $y^5 \times y = y^5$

Forgot the implied exponent of 1. Correct: y^6

c $a^2 \times a^2 = 2a^4$

Added the bases. Correct: a^4

d $b^3 \times b^4 = b^{34}$

Wrote the exponents next to each other. Correct: b^7

e $m^5 \times n^3 = mn^8$

Bases different, cannot use index laws. Correct simplification is m^5n^3

f $p^2 \times p^3 = 2p^5$

There is no 2; just keep the base. Correct: p^5

g $w^6 \times w^2 = w^4$

Subtracted instead of added. Correct: w^8

14 Find the value of x in these equations.

a $3^x \times 3^x = 3^{10}$

$$3^{2x} = 3^{10}$$

$$2x = 10$$

$$x = 5$$

b $5^{x+1} \times 5^{2x} = 5^{13}$

$$4$$

c $a^x \times a^{4x} \times a^2 = a^{32}$

$$6$$

d $7^{x-5} \times 49 = 7^6$

$$9$$

e $m^{2x} \times m^{x-2} = m^x \times m^4$

$$3$$

Index Law of Division

Review

- ☐ Simplify fractions by cancelling common factors

a

$$\frac{5 \times 5 \times 5}{5}$$

b

$$\frac{7 \times 7 \times 7 \times 7 \times 7}{7 \times 7}$$

c

$$\frac{7}{7 \times 7 \times 7}$$

d

$$\frac{2 \times 2}{2 \times 2 \times 2}$$

- ☐ Recall index law of division for numerical bases

a $5^7 \div 5^3$

b $3^4 \div 3$

c $\frac{7^5}{7^2}$

d $\frac{2^9}{2^3}$

Index Law of Division

$$a^m \div a^n = a^{m-n}$$

Simplify using index law of division

$$x^7 \div x^3$$

$$= x^4$$

$$x^{43} \div x^{40}$$

$$= x^3$$

$$y^{953} \div y^{900}$$

$$= y^{53}$$

Simplify the following expressions

a $x^6 \div x^2$

b $a^7 \div a^3$

c $y^8 \div y$

d $a^9 \div a^2 \div a^4$

e $y^5 \div y^5$

f $\frac{x^7}{x^2}$

g $\frac{y^9}{y^4}$

h $x^5 - x^3$

i $x^3 \div y^2$

① Index Law of Division

When the **numerator** has a **higher power**, find the difference of the powers, and the base is in the numerator.

$$\frac{x^{11}}{x^2} \div \frac{x^2}{x^2} = \frac{x^9}{1} = x^9$$

When the **denominator** has a **higher power**, find the difference of the powers, and the base is in the denominator.

$$\frac{x^2}{x^{11}} \div \frac{x^2}{x^2} = \frac{1}{x^9}$$

② Apply Index law of division

$$\frac{x^5}{y^3}$$

Law doesn't apply

$$\frac{x^9}{x^2} = x^7$$

$$\frac{x^2}{x^7} = \frac{1}{x^5}$$

$$\frac{x^{20}}{x^7} = x^{13}$$

$$\frac{x^{20}}{x^{70}} = \frac{1}{x^{50}}$$

Simplify the following expressions

a $\frac{x^6}{x^2}$

b $\frac{x^2}{x^6}$

c $\frac{x^5}{x^{11}}$

d $\frac{x^{11}}{x^5}$

e $\frac{y^{30}}{y^{26}}$

f $\frac{m^3}{m^{10}}$

g $\frac{p^9}{p^2}$

h $\frac{m^2}{m^{15}}$

💡 Key Ideas

1. When dividing powers with the same base, the powers.
2. $x^m \div x^n = \dots\dots\dots$
3. Fractions also represent so $\frac{x^m}{x^n} = \dots\dots\dots$
4. $\frac{x^{20}}{x^4} = \dots\dots\dots$ and $\frac{x^4}{x^{20}} = \dots\dots\dots$
5. The law only applies when the base is the in both terms

1 Fill in the boxes.

$$\text{a } 5^5 \div 5^3 = \frac{5 \times \square \times \square \times \square \times \square}{5 \times \square \times \square} = 5^{\square}$$

$$\text{b } 9^4 \div 9^2 = \frac{9 \times \square \times \square \times \square}{9 \times \square} = 9^{\square}$$

$$\text{c } a^6 \div a^2 = \frac{a \times \square \times \square \times \square \times \square \times \square}{\square \times \square} = a^{\square}$$

$$\text{d } \frac{n^6}{n^3} = \frac{n \times \square \times \square \times \square \times \square \times \square}{\square \times \square \times \square} = n^{\square}$$

2 Write as a single power.

$$\text{a } x^5 \div x^2$$

x^3

$$\text{d } x^8 \div x^3$$

x^5

$$\text{g } m^{12} \div m^7$$

m^5

$$\text{b } x^6 \div x^2$$

x^4

$$\text{e } a^9 \div a^4$$

a^5

$$\text{h } n^{15} \div n^6$$

n^9

$$\text{c } x^7 \div x^2$$

x^5

$$\text{f } b^{10} \div b^5$$

b^5

$$\text{i } p^{20} \div p^{15}$$

p^5

3 Write as a single power.

$$\text{a } \frac{x^5}{x^2}$$

x^3

$$\text{d } \frac{c^{10}}{c^4}$$

c^6

$$\text{g } \frac{n^{20}}{n^5}$$

n^{15}

$$\text{b } \frac{a^7}{a^3}$$

a^4

$$\text{e } \frac{h^8}{h^3}$$

h^5

$$\text{h } \frac{p^7}{p}$$

p^6

$$\text{c } \frac{b^9}{b^2}$$

b^7

$$\text{f } \frac{m^{15}}{m^{10}}$$

m^5

$$\text{i } \frac{q^{100}}{q^{50}}$$

q^{50}

4 Write each of these with a single power.

$$\text{a } n^6 \div n^2$$

n^4

$$\text{b } x^7 \div x^3$$

x^4

$$\text{c } 7^5 \div 7^3$$

7^2

$$\text{d } q^4 \div q^3$$

q

$$\text{e } 5^4 \div 5$$

5^3

$$\text{f } y^{50} \div y^2$$

y^{48}

$$\text{g } e^4 \div e$$

e^3

$$\text{h } a^{14} \div a^5$$

a^9

$$\text{i } \frac{x^6}{x^2}$$

x^4

$$\text{j } \frac{n^7}{n}$$

n^6

$$\text{k } \frac{7^9}{7^5}$$

7^4

$$\text{l } \frac{r^{30}}{r^{10}}$$

r^{20}

- 5 For each question, write **N** if the numerator has the higher power and **D** if the denominator has the higher power.

a $\frac{x^6}{x^2}$ N	b $\frac{p^2}{p^9}$ D	c $\frac{c^{12}}{c^3}$ N	d $\frac{t^4}{t^9}$ D
e $\frac{m^3}{m^{10}}$ D	f $\frac{y^{30}}{y^{26}}$ N	g $\frac{h^5}{h^{50}}$ D	h $\frac{w^{11}}{w}$ N
i $\frac{a^{15}}{a^4}$ N	j $\frac{b^8}{b^{20}}$ D	k $\frac{r^9}{r^4}$ N	l $\frac{u^2}{u^{12}}$ D

- 6 Explain why these two cards give the same answer.

A $\frac{g^8}{g^4}$

B $g^8 \div g^4$

The fraction bar represents division

Simplify each expression

a i $d^6 \div d^3$

ii $d^6 \div d^4$

iii $d^6 \div d^5$

d^3

d^2

d

b i $\frac{y^3}{y}$

ii $\frac{y^4}{y}$

iii $\frac{y^5}{y}$

y^2

y^3

y^4

- 7 Explain why you can't simplify $\frac{g^7}{m^4}$

The bases are not the same.

- 8 Write each in the form $\frac{1}{\text{base}^n}$

a $\frac{x^2}{x^5}$

d $\frac{c^4}{c^{10}}$

g $\frac{n^5}{n^{20}}$

$\frac{1}{x^3}$

$\frac{1}{c^6}$

$\frac{1}{n^{15}}$

b $\frac{a^3}{a^7}$

e $\frac{h^3}{h^8}$

h $\frac{p}{p^7}$

$\frac{1}{a^4}$

$\frac{1}{h^5}$

$\frac{1}{p^6}$

c $\frac{b^2}{b^9}$

f $\frac{m^{10}}{m^{15}}$

i $\frac{q^{50}}{q^{100}}$

$\frac{1}{b^7}$

$\frac{1}{m^5}$

$\frac{1}{q^{50}}$

9 Look at the example and then answer the questions.

Simplify these.

$$\frac{h^9}{h^4} = h^5$$

$$\frac{h^3}{h^7} = \frac{1}{h^4}$$

a $\frac{n^3}{n^5} = \frac{\boxed{}}{\boxed{}}$

$\frac{1}{n^2}$

b $\frac{n^5}{n^3} = \boxed{}$

n^2

c $\frac{a^7}{a^2} = \boxed{}$

a^5

d $\frac{a^2}{a^7} = \frac{\boxed{}}{\boxed{}}$

$\frac{1}{a^5}$

e $\frac{x}{x^4} = \frac{\boxed{}}{\boxed{}}$

$\frac{1}{x^3}$

f $\frac{x^4}{x} = \boxed{}$

x^3

g $\frac{y^5}{y^4} = \boxed{}$

y

h $\frac{y^4}{y^5} = \boxed{}$

$\frac{1}{y}$

i $\frac{b^2}{b^9} = \boxed{}$

$\frac{1}{b^7}$

j $\frac{b^9}{b^2} = \boxed{}$

b^7

k $\frac{c^5}{c^8} = \boxed{}$

$\frac{1}{c^3}$

l $\frac{k^2}{k^7} = \boxed{}$

$\frac{1}{k^5}$

10 Simplify.

a $\frac{a^8}{a^3} = \boxed{}$

a^5

b $\frac{b^4}{b^{11}} = \boxed{}$

$\frac{1}{b^7}$

c $c^{15} \div c^5 = \boxed{}$

c^{10}

d $\frac{m^7}{m^2} = \boxed{}$

m^5

e $\frac{n^3}{n^{12}} = \boxed{}$

$\frac{1}{n^9}$

f $\frac{q^6}{q^9} = \boxed{}$

$\frac{1}{q^3}$

g $s^{100} \div s^{99} = \boxed{}$

s

h $\frac{t}{t^5} = \boxed{}$

$\frac{1}{t^4}$

11 Simplify each pair.

a $\frac{x^7}{x^2}$ and $\frac{x^2}{x^7}$

$x^5, \frac{1}{x^5}$

b $\frac{a^5}{a^3}$ and $\frac{a^3}{a^5}$

$a^2, \frac{1}{a^2}$

c $\frac{m^4}{m^{10}}$ and $\frac{m^{10}}{m^4}$

$\frac{1}{m^6}, m^6$

d $\frac{n^{12}}{n^7}$ and $\frac{n^7}{n^{12}}$

$n^5, \frac{1}{n^5}$

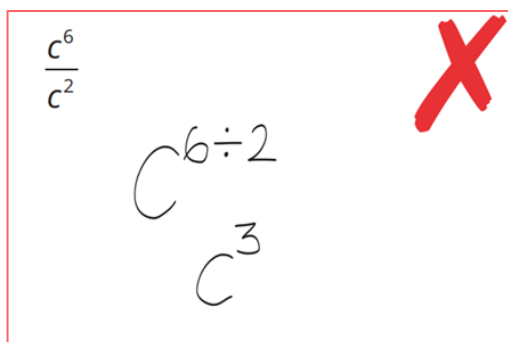
e $\frac{p^{15}}{p^{20}}$ and $\frac{p^{20}}{p^{15}}$

$\frac{1}{p^5}, p^5$

f $\frac{q}{q^9}$ and $\frac{q^9}{q}$

$\frac{1}{q^8}, q^8$

12 Read the example carefully and answer the questions.



This student divided her exponents to get her answer. What operation should they have used to get the correct answer?

subtract

Simplify:

$$\frac{b^6}{b^3}$$

b^3

13 Ally says:



$$8^{17} \div 8^{15} = 1^2$$

Explain why she is incorrect and state the correct answer.

The base number does not change.

8^2

14 Every answer is **incorrect**, state the error and write the correct answer.

a $\frac{a^7}{a^3} = a^{\frac{7}{3}}$

Divided the powers. Correct: a^4

b $b^8 \div b^5 = 1^3$

Divided the base too. Correct: b^3

c $\frac{c^9}{c^4} = c^{13}$

Added instead of subtracted. Correct: c^5

d $\frac{d^3}{d^{10}} = d^7$

Numerator is smaller, answer should have base on the bottom. Correct: $\frac{1}{d^7}$

e $f^{10} \div f = f^{10}$

Forgot the implied power of 1. Correct: f^9

15 Work out the missing powers.

a $x^8 \div x^{\boxed{}} = x^5$

b $4^{\boxed{}} \div 4^2 = 4^6$

c $y^{10} \div y^{\boxed{}} = y^{10}$

d $q^{\boxed{}} \div q = q^5$

3

8

0

6

16 Simplify these expressions.

a $7^{5a} \div 7^{2a}$

b $y^{5x+3} \div y^{2x}$

c $y^{2m+3} \div y^{m+5}$

7^{3a}

y^{3x+3}

y^{m-2}

17 Evaluate without a calculator.

a $7^7 \div 7^5$

b $10^6 \div 10^5$

c $13^{11} \div 13^9$

d $2^{20} \div 2^{17}$

e $7 \times 31^{16} \div 31^{15}$

f $3 \times 50^{200} \div 50^{198}$

49

10

169

8

217

7500

18 Consider this information:

$2^9 = 512$

$2^{11} = 2048$

$2^{20} = 1\,048\,576$

Use this information to work out the answers to these calculations.

a $1\,048\,576 \div 2048$

b $1\,048\,576 \div 512$

512

2048

Index Law of Power of a Power

Review

- ☐ Simplify expressions using the index law of multiplication

a $x^3 \times x^3$

b $x^3 \times x^3 \times x^3$

c $x^3 \times x^3 \times x^3 \times x^3$

d $(y^5)^2$

- ☐ Write out power of a power in expanded form

a $(x^3)^4$

b $(x^5)^2$

Index Law of Power of a Power

$$(a^m)^n = a^{m \times n}$$

Simplify using index law of power of a power

$$(x^3)^4$$

$$= x^{12}$$

$$(x^3)^5$$

$$= x^{15}$$

$$(a^{20})^6$$

$$= a^{120}$$

Simplify the following expressions

a $(x^2)^3$

b $(y^3)^5$

c $(a^4)^2$

d $(b^2)^4$

e $(n^{10})^2$

f $(x^2)^3 \times x^4$

② **Identify** whether to add to multiply the powers

$x^3 \times x^4$ Add	$(x^3)^4$ Multiply	$(a^5)^2$ Multiply	$(m^{10})^3$ Multiply	$x^3 \times x^4 \times x^5$ Add
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Identify whether to add the powers or multiply the powers.

a $(y^4)^3$	b $x^2 \times x^5$	c $(a^7)^2$	d $(c^5)^3$
e $b^4 \times b^9$	f $m^6 \times m \times m^2$	g $x^2 x^5$	h $(m^2)^3$

💡 **Key Ideas**

1. When two terms are multiplied, e.g. $x^3 \times x^4$, we the powers.
2. When a power is raised to another power, e.g. $(x^3)^4$, we the powers.

1 For each, write + if you add the powers, and $\times$ if you multiply the powers.

a $(x^3)^4$	b $b^4 \times b^9$	c $(n^7)^3$	d $r^4 \times r^4$
$\times$	$+$	$\times$	$+$
e $x^3 \times x^4$	f $(c^3)^5$	g $p^8 \times p^2$	h $(s^2)^9$
$+$	$\times$	$+$	$\times$
i $(a^5)^2$	j $m^6 \times m \times m^2$	k $(q^{10})^4$	l $(t^6)^6$
$\times$	$+$	$\times$	$\times$

2 Complete the following.

- a** $(4^2)^3 = 4^2 \times 4^2 \times 4^2 = \boxed{}$ 4^6
- b** $(5^3)^4 = 5^3 \times \boxed{} \times \boxed{} \times \boxed{} = \boxed{}$ 5^{12}
- c** $(d^4)^3 \equiv \boxed{} \times \boxed{} \times \boxed{} \equiv \boxed{}$ d^{12}
- d** $(v^5)^6 \equiv \boxed{} \times \boxed{} \times \boxed{} \times \boxed{} \times \boxed{} \times \boxed{} \equiv \boxed{}$ v^{30}

3 Write each of these as a single power.

a i $(5^3)^2$	ii $(5^3)^3$	iii $(5^3)^4$	iv $(5^3)^5$	v $(5^3)^6$
5^6	5^9	5^{12}	5^{15}	5^{18}
b i $(e^2)^5$	ii $(e^3)^5$	iii $(e^4)^5$	iv $(e^5)^5$	v $(e^6)^5$
e^{10}	e^{15}	e^{20}	e^{25}	e^{30}

4 Simplify

a $(x^7)^4$	b $(p^4)^2$	c $(v^8)^3$	d $(d^6)^3$
x^{28}	p^8	v^{24}	d^{18}
e $(t^4)^{10}$	f $(x^2)^5$	g $(p^6)^5$	h $(u^3)^4$
t^{40}	x^{10}	p^{30}	u^{12}
i $(4^2)^4$	j $(5^3)^9$	k $(22^3)^5$	l $(176.4^{20})^{50}$
4^8	5^{27}	22^{15}	$(176.4)^{1000}$

5 Simplify.

a $(x^3)^4$	b $b^4 \times b^9$	c $(n^7)^3$	d $r^4 \times r^4$
x^{12}	b^{13}	n^{21}	r^8
e $x^3 \times x^4$	f $(c^3)^5$	g $p^8 \times p^2$	h $(s^2)^9$
x^7	c^{15}	p^{10}	s^{18}
i $(a^5)^2$	j $m^6 \times m \times m^2$	k $(q^{10})^4$	l $(t^6)^6$
a^{10}	m^9	q^{40}	t^{36}

6 Simplify

a	$(6^2)^5 =$	j	$(3^2)^{\frac{1}{2}} =$
b	$(6^3)^5 =$	k	$(3^{\frac{1}{2}})^2 =$
c	$(6^4)^5 =$	l	$(3^{\frac{1}{2}})^{\frac{1}{2}} =$
d	$(2^4)^5 =$	m	$(x^2)^2 =$
e	$(2^5)^4 =$	n	$(x^2)^7 =$
f	$(2^5)^0 =$	o	$(2^7)^x =$
g	$(2^5)^{-1} =$	p	$(7^x)^2 =$
h	$(2^{-5})^{-1} =$	q	$(x^2)^5 =$
i	$(3^{-5})^{-1} =$	r	$(x^2)^a =$

7 Simplify

a	$(p^{\frac{1}{9}})^{\frac{2}{5}}$	b	$(h^{\frac{3}{4}})^{\frac{9}{10}}$
----------	-----------------------------------	----------	------------------------------------

8 Simplify

a	$(x^2)^3 \times x^4$	b	$(y^3)^2 \times y^5$	c	$(a^4)^2 \times a$
d	$b \times (b^2)^4$	e	$(m^3)^3 \times m^3$	f	$(n^5)^2 \times n^2 \times n$
g	$(p^2)^4 \times (p^3)^2$	h	$(q^3)^2 \times (q^2)^3$		

9 Without simplifying, explain why these expressions are equivalent.

$(r^{273})^{95}$

$(r^{95})^{273}$

You can multiply in any order and the product will be the same.

10 Work out the missing numbers.

a $(2^3)^{\square} = 2^{12}$

4

b $(a^{\square})^6 \equiv a^{42}$

7

c $(g^4)^{\square} \equiv g^{40}$

10

11 Work out the missing numbers.

a $(x^2)^3 \times x^{\square} = x^{11}$

5

b $(x^4)^{\square} \times (x^3)^2 = x^{14}$

2

12 Every answer is **incorrect**, state the error and write the correct answer.

a $(x^3)^4 = x^7$

Added instead of multiplied. Correct: x^{12}

b $(a^5)^2 = a^{25}$

Squared instead of multiplied by 2. Correct: a^{10}

c $(b^2)^6 = 2b^{12}$

There is no 2 in front. Correct: b^{12}

d $(m^{10})^3 = m^{13}$

Added instead of multiplied. Correct: m^{30}

Index Law of Zero

Review

☐ Recall index law of division

a $x^5 \div x^2$

b $n^6 \div n^5$

c $\frac{y^7}{y^2}$

d $\frac{y^7}{y^7}$

☐ Divide a number by itself

a $7 \div 7$

b $\frac{7}{7}$

c $x^3 \div x^3$

d $\frac{y^7}{y^7}$

Index Law of Zero

$$x^0 = 1$$

Simplify using index law of zero

An important question to ask simplifying is 'what does the zero apply to?'

$$\begin{aligned} 5x^0 \\ = 5 \times 1 \\ = 5 \end{aligned}$$

$$\begin{aligned} (5x)^0 \\ = 1 \end{aligned}$$

$$\begin{aligned} (3ab)^0 \\ = 1 \end{aligned}$$

$$\begin{aligned} x(y^2)^0 \\ = x \times 1 \\ = x \end{aligned}$$

$$\begin{aligned} 3 + a^0 \\ = 3 + 1 \\ = 3 \end{aligned}$$

Simplify the following expressions

a $2x^0$

b $(2x)^0$

c $(4ab)^0$

d $4ab^0$

e $6 - p^0$

f $7x^0$

g $2x(3yz)^0$

h $x + 3a^0$

Key Ideas

- Any base raised to the power zero equals
- In the expression $5x^0$, the power of zero applies only to
In the expression $(5x)^0$, the power of zero applies to

1 Simplify these expressions using the power of zero.

a 3^0	b 25^0	c $(-5)^0$
1	1	1
d 7^0	e 100^0	f $\left(\frac{1}{2}\right)^0$
1	1	1
g 10^0	h $(-2)^0$	i 0.7^0
1	1	1

2

a Explain why $\frac{3^2}{3^2} = 1$

Anything divided by itself is 1

b Using your answer to part **a**, explain why $3^0 = 1$

$$3^2 \div 3^2 = 3^0 = 1$$

3 Simplify these expressions using the power of zero.

a $2x^0$	b $10m^0$	c $4 + b^0$
2	10	5
d $5y^0$	e $-3p^0$	f $9 - c^0$
5	-3	8
g $7a^0$	h $-6q^0$	i $3n^0 + 1$
7	-6	4

4 Simplify these expressions using the power of zero.

a $(2x)^0$	b $(10m)^0$	c $(4b)^0$
1	1	1
d $(5y)^0$	e $(-3p)^0$	f $(9c)^0$
1	1	1
g $(7a)^0$	h $(3n)^0 + 1$	i $(5h)^0 - 2$
1	2	-1

5 Simplify these expressions using the power of zero.

a $(xyz)^0$	b $(5mn)^0$	c $a(bc)^0$	d $3(xy)^0$
1	1	a	3
e $7m(np)^0$	f $-4(rs)^0$	g $b(bc)^0$	h $-(2b)^0$
7m	-4	b	-1

6 Simplify these expressions using the power of zero.

a $2^0 =$

h $6^0 =$

b $3^0 =$

i $(-6)^0 =$

c $4^0 =$

j $-(6)^0 =$

d $5^0 =$

k $\left(\frac{1}{6}\right)^0 =$

e $0^5 =$

l $(0.6)^0 =$

f $0^4 =$

m $(6p)^0 =$

g $0^0 =$

n $6p^0 =$

undefined

7 Simplify these pairs of expressions using the power of zero.

a $4x^0$ and $(4x)^0$

b $6y^0$ and $(6y)^0$

c $-2a^0$ and $(-2a)^0$

d $5m^0 + 3$ and $(5m)^0 + 3$

e $2x^0 + 2x^0$ and $2x^0 + (2x)^0$

f $-7a^0$ and $(-7a)^0$

8 Evaluate these. Example:

$$5m^0 + 7$$

$$= 5 \times 1 + 7 = 12$$

a $2v^0$
 $= 2 \times 1 =$

b $8h^0$

c $h^0 + 7$

d $7h^0 - 4$

e x^0

f $2x^0$

g $2x^0 + 2$

h $x^0 + 2x^0$

i n^0

j 4^0

k $4n^0$

l $(4n)^0$

m $4n^0 + 4$

n $(n^0 + 4)^2$

o $4n^0 + (4n)^0$

p $4^0 n$

9 Every answer is **incorrect**, state the error and write the correct answer.

a $5x^0 = 0$

The 0 only applies to x , so $x^0 = 1$ and the 5 stays. Correct: 5

b $(7a)^0 = 7$

The brackets mean the 0 applies to everything inside. Correct: 1

c $3a^0 = 1$

The 0 applies only to a , so $a^0 = 1$, then 3×1 . Correct: 3

d $(2bc)^0 = 2$

The brackets mean the 0 applies to everything inside. Correct: 1

e $4 + m^0 = 4$

$m^0 = 1$, not 0. Correct: 5

f $-2y^0 = 2$

$y^0 = 1$, then $-2 \times 1 = -2$. Correct: -2

Index Laws with Coefficients

Review

☐ Recall index laws

a $x^3 \times x^5$

b $a^2 \times a^4$

c $\frac{x^8}{x^3}$

d $\frac{a^6}{a^2}$

e $y \times y^6$

f $\frac{m^4}{m^7}$

g $(x^5)^3$

h $(x^2)^5$

☐ Simplify algebraic expressions

a $3x \times 5y$

b $4x \times 5y$

c $12x \div 6y$

d $16x \div 4$

e $\frac{16x}{8}$

f $\frac{16}{8x}$

Identify coefficients and variables

In algebra, **variables** are letters and **coefficients** are the numbers multiplied by them. We work on each **separately**.

$4x^3 \times 3x^5$

Coefficients: 4×3

Variables: $x^3 \times x^5$

$2y^4 \times 5y^2$

Coefficients: 2×5

Variables: $y^4 \times y^2$

$\frac{12a^7}{4a^3}$

Coefficients: $\frac{12}{4}$

Variables: $\frac{a^7}{a^3}$

Identify the coefficients and the variables:

a $6x^2 \times 2x^5$

Coefficients:

Variables:

b $\frac{10y^6}{5y^2}$

Coefficients:

Variables:

c $3a^4 \times 4a$

Coefficients:

Variables:

d $\frac{8m^9}{2m^3}$

Coefficients:

Variables:

Example Apply index laws to simplify algebraic expressions

$$4x^3 \times 3x^5$$

$$= 12x^8$$

$$2x^4 \times 5x^3$$

$$= 10x^7$$

Guided Practice

a $3x^2 \times 4x^5$

b $2x^3 \times 4x^4$

c $-5x^2 \times 3x^3$

d $3x^4 \times 2x^3 \times 5x^{10}$

$$12x^7$$

$$8x^7$$

$$-15x^5$$

$$30x^{17}$$

Example Apply index laws to simplify algebraic expressions with coefficients

$$\frac{8x^5}{6x^3}$$

$$= \frac{4x^2}{3}$$

$$\frac{8x^3}{6x^5}$$

$$= \frac{4}{3x^2}$$

$$\frac{12x^8}{4x^3}$$

$$= 3x^5$$

Guided Practice

a $\frac{8b^5}{12b^3}$

b $\frac{3m^2}{12m^7}$

c $-9m^6 \div 3m^2$

d $2x^3 \times 6x^4 \div 4x^2$

$$\frac{2b^2}{3}$$

$$\frac{1}{4m^5}$$

$$-3m^4$$

$$3x^5$$

Key Ideas

1. When multiplying terms like $4x^3 \times 3x^5$, multiply the and the indices
2. When dividing terms like $12x^8 \div 4x^3$, divide the and the indices
3. The coefficients and the variables are handled

1 Simplify each of these expressions.

a i $x^4 \times x^2$	ii $-x^4 \times x^2$	iii $-x^4 \times -x^2$	
x^6	$-x^6$	x^6	
b i $2m^3 \times m^2$	ii $m^3 \times 2m^2$	iii $3m^3 \times 2m^2$	iv $5m^3 \times 4m^2$
$2m^5$	$2m^5$	$6m^5$	$20m^5$

2 Simplify.

a $5x^2 \times 2x^4$	b $2b^6 \times 5b^2$	c $5n^4 \times 4n^4$
$10x^6$	$10b^8$	$20n^8$
d $3a^4 \times 4a$	e $4c^3 \times 3c^7$	f $2p^5 \times 7p^5$
$12a^5$	$12c^{10}$	$14p^{10}$
g $6y^3 \times 2y^5$	h $8h^2 \times 2h^3$	i $6q^{10} \times 2q^2$
$12y^8$	$16h^5$	$12q^{12}$
j $7m \times 3m^4$	k $9k \times 2k^9$	l $3r^8 \times 5r$
$21m^5$	$18k^{10}$	$15r^9$

3 Simplify.

a $-3x^2 \times 4x^3$	b $-6y^2 \times -3y^5$	c $7p^3 \times -p^5$
$-12x^5$	$18y^7$	$-7p^8$
d $5a^4 \times -2a^3$	e $-4m \times 2m^6$	f $-2q^4 \times -5q^4$
$-10a^7$	$-8m^7$	$10q^8$

4 Simplify each of the following.

a $x^8 \div x^2$	b $-x^8 \div x^2$	c $-x^8 \div -x^2$
x^6	$-x^6$	x^6
d $10m^6 \div m^2$	e $10m^6 \div 2$	f $10m^6 \div 2m^2$
$10m^4$	$5m^6$	$5m^4$

5 Simplify.

a $\frac{10x^7}{2x^3}$	b $\frac{20m^8}{4m^2}$	c $\frac{18b^{10}}{6b^4}$	d $\frac{30p^{12}}{6p^5}$
$5x^4$	$5m^6$	$3b^6$	$5p^7$
e $\frac{12a^9}{3a^5}$	f $\frac{15y^6}{5y^2}$	g $\frac{24c^7}{8c}$	h $\frac{16q^5}{2q^5}$
$4a^4$	$3y^4$	$3c^6$	8

6 Study the example and simplify these.

$$\begin{aligned}(3x^0)^3 &= (3 \times 1)^3 \\ &= (3)^3 \\ &= 27\end{aligned}$$

a $(5n^0)^4 = (\square \times \square)^4$
 $= \square^4 = \square$
625

b $(8p^0)^3 = (\square \times \square)^3$
 $= \square^3 = \square$
512

c $(2h^0)^9 = (\square \times \square)^9$
 $= \square^9 = \square$
512

d $(2h^0)^3$
8

e $(8k^0)^2$
64

f $(3x^0y^0)^4$
81

g $(2h^3)^0$
1

h $(8k^2)^0$
1

i $(3x^2y^4)^0$
1

7 Simplify these expressions.

a $x^0 \times y^0$
1
 $x + 1$

b $x^0 + y^0$
2
3

c $a^0 - b^0$
0
0

d $p \times q^0$
 p
 $6p$

e $x + y^0$
 $x + 1$

f $2x^0 + y^0$
3

g $(2x)^0 - z^0$
0

h $2p \times 3q^0$
 $6p$

8 Simplify the following.

a $\frac{6x^2}{x}$
 $6x$

b $\frac{x^2}{6x}$
 $\frac{x}{6}$

c $\frac{x}{6x^2}$
 $\frac{1}{6x}$

d $\frac{3x}{6x^2}$
 $\frac{1}{2x}$

e $\frac{6x^2}{3x}$
 $2x$

f $\frac{6x^2}{3x^2}$
2

g $\frac{9x^2}{6x}$
 $\frac{3x}{2}$

h $\frac{9x}{6x^2}$
 $\frac{3}{2x}$

i $15 \div 15a^2$
 $\frac{1}{a^2}$

j $2y \div 8y^2$
 $\frac{1}{4y}$

k $8y^3 \div 6y$
 $\frac{4y^2}{3}$

l $9y^2 \div 12y^5$
 $\frac{3}{4y^3}$

9 Simplify.

a $\frac{4x^2}{8x^5}$

$\frac{1}{2x^3}$

b $\frac{3m^2}{15m^8}$

$\frac{1}{5m^6}$

c $\frac{7p}{14p^6}$

$\frac{1}{2p^5}$

d $\frac{6a^3}{12a^9}$

$\frac{1}{2a^6}$

e $\frac{5y^4}{20y^{10}}$

$\frac{1}{4y^6}$

f $\frac{2b^3}{16b^{11}}$

$\frac{1}{8b^8}$

10 Simplify these pairs of expressions.

a i $\frac{g^8}{g^2}$ ii $\frac{g^2}{g^8}$

$g^6, \frac{1}{g^6}$

b i $\frac{8k^3}{2k}$ ii $\frac{8k}{2k^3}$

$4k^2, \frac{4}{k^2}$

c i $\frac{12p^7}{6p^2}$ ii $\frac{6p^2}{12p^7}$

$2p^5, \frac{1}{2p^5}$

d i $\frac{12x^5}{9x^4}$ ii $\frac{9x^4}{12x^5}$

$\frac{4x}{3}, \frac{3}{4x}$

11 Simplify.

a $\frac{6x^5 \times 2x^3}{4x^4}$

$3x^4$

b $2x^3 \times 6x^4 \div 4x^2$

$3x^5$

c $5y^2 \times 4y^3 \times 2y$

$40y^6$

d $\frac{3a^2 \times 4a^6}{2a^3}$

$6a^5$

e $\frac{8m^9}{2m^3 \times m^2}$

$4m^4$

f $\frac{10p^4 \times 3p^5}{6p^3}$

$5p^6$

12 Tammy is trying to simplify $2q^4 \times 4q^5$



The answer is $6q^9$

Explain why she is incorrect and state the correct answer.

We multiply the non-base numbers. $8q^9$

13 Every answer is **incorrect**, state the error and write the correct answer.

a $2x^3 \times 3x^5 = 5x^8$

Added the coefficients instead of multiplying. Correct: $6x^8$

b $4a^2 \times 3a^5 = 12a^{10}$

Multiplied the exponents instead of adding. Correct: $12a^7$

c $\frac{8y^6}{2y^2} = 4y^3$

Did not subtract the exponents correctly. Correct: $4y^4$

d $\frac{10m^4}{10m^4} = 0$

Everything cancels, but the answer should be 1.

e $3p^2 \times 5p^3 = 8p^5$

Added the coefficients. Correct: $15p^5$

f $-2x^4 \times 3x^2 = 6x^6$

Forgot the negative sign. Correct: $-6x^6$

Index Laws with More than One Variable

Review

☐ Recall index laws with coefficients

a $3x^2 \times 2x^3$

b $4x^4 \times 2x^3$

c $6x^5 \div 2x^2$

☐ Recall implied index of 1

a $x \times x^2$

b $y^5 \div y$

c $m^5 \times m$

☐ **Simplify** by applying index laws to each variable separately

When applying index laws, we apply them to each variable separately

$$\begin{aligned} x^2y^4 \times x^3y^7 \\ = x^5y^{11} \end{aligned}$$

$$\begin{aligned} x^2y^4 \times x^3y \\ = x^5y^5 \end{aligned}$$

$$\begin{aligned} \frac{x^5y^7}{x^{11}y^2} \\ = \frac{y^5}{x^6} \end{aligned}$$

Simplify these expressions

a $x^3y^2 \times x^2y^4$

b $x^2y^3 \times yx^5$

c $a^3b \times ab^5$

d $\frac{x^9y^6}{xy^2}$

e $\frac{x^9y^6}{x^{12}y^2}$

f $a^2b^3c \times ab^5$

 **Example** Apply index laws to simplify expressions with more than one variable

$$3x^2y^4 \times 2x^3y$$

$$= 6x^5y^5$$

$$\frac{9x^5y^2}{6x^2y^7}$$

$$= \frac{3x^3}{2y^5}$$

Guided Practice

Simplify

a $6x^3y^7 \times 2x^5y^2$

b $4c^4d \times 2c^2$

c $\frac{12x^2y^{11}}{8x^6y}$

d $\frac{2x^5y^3}{14x^6y}$

$$12x^8y^9$$

$$8c^6d$$

$$\frac{3y^{10}}{2x^4}$$

$$\frac{y^2}{7x}$$

Key Ideas

1. When multiplying expressions with multiple variables, apply the index law to each variable

.....

1 Simplify:

a $a^2b^3 \times a^4b$

a^6b^4

b $x^3y^2 \times x^4y^6$

x^7y^8

c $u^6v^2 \times u^2v^7$

u^8v^9

d $m^5n^2 \times mn^4$

m^6n^6

e $h^2k^5 \times h^3k^2$

h^5k^7

f $c^7d^3 \times cd^9$

c^8d^{12}

g $pq^3 \times p^2q^5$

p^3q^8

h $r^4s \times rs^3$

r^5s^4

i $w^4y^4 \times w^4y^4$

w^8y^8

2 Simplify:

a $\frac{a^7b^5}{a^3b^2}$

a^4b^3

b $\frac{x^{10}y^4}{x^4y}$

x^6y^3

c $\frac{r^5s^7}{rs^2}$

r^4s^5

d $\frac{m^8n^6}{m^5n^3}$

m^3n^3

e $\frac{p^6q^9}{p^2q^5}$

p^4q^4

f $\frac{c^9d^{12}}{c^2d^4}$

c^7d^8

3 Simplify these expressions.

a $f^8g^3 \div f^2g^2$

f^6g

b $f^8g^2 \div f^2g^3$

$\frac{f^6}{g}$

c $f^2g^3 \div f^8g^2$

$\frac{g}{f^6}$

d $f^2g^2 \div f^8g^3$

$\frac{1}{f^6g}$

4 Simplify:

a $\frac{x^7y^3}{x^2y^8}$

$\frac{x^5}{y^5}$

b $\frac{m^4n^6}{m^{10}n^2}$

$\frac{n^4}{m^6}$

c $\frac{r^2s^7}{r^9s^3}$

$\frac{s^4}{r^7}$

d $\frac{a^3b^9}{a^7b^5}$

$\frac{b^4}{a^4}$

e $\frac{p^8q}{p^2q^4}$

$\frac{p^6}{q^3}$

5 Simplify. Example:

$$\frac{b^2 e^2}{b^3 e^4} = \frac{1}{be^2} \quad \frac{b^5 e^5}{be^3} = b^4 e^2$$

a $\frac{ab^9}{b^7} = \boxed{}$ ab^2	b $\frac{b^9}{ab^7} = \frac{\boxed{}}{\boxed{}}$ $\frac{b^2}{a}$	c $\frac{x^2 y^9}{y^7} = \boxed{}$ $x^2 y^2$	d $\frac{x^2 y^9}{xy^7} = \boxed{}$ xy^2
e $\frac{a^2 n^9}{a^2 n^8} = \boxed{}$ n	f $\frac{a^5 n^3}{a^2 n^9} = \frac{\boxed{}}{\boxed{}}$ $\frac{a^3}{n^6}$	g $\frac{bc}{b^2 c} = \frac{\boxed{}}{\boxed{}}$ $\frac{1}{b}$	h $\frac{c^5 b^2}{b^2 c^3} = \boxed{}$ c^2
i $\frac{s^4 v}{v^2 s^2} = \boxed{}$ $\frac{s^2}{v}$	j $\frac{xy^2}{x^2 y} = \boxed{}$ $\frac{y}{x}$	k $\frac{a^4 e}{e^2 a^3} = \boxed{}$ $\frac{a}{e}$	l $\frac{c^2 f^2}{cf^7} = \boxed{}$ $\frac{c}{f^5}$

6 Simplify the following expressions.

a $f^3 \times h^3 \times f^4 \times h^4$ $f^7 h^7$	b $3m^2 n \times m^8$ $3m^{10} n$	c $a^2 b^3 \times a^3 b^2$ $a^5 b^5$
d $x^5 y^2 \times y^4 x^3$ $x^8 y^6$	e $m^2 n^3 \times mn^4$ $m^3 n^7$	f $3c^{12} d^5 \times c^4 d^5$ $3c^{16} d^{10}$
g $6gh^4 \times 2h^3$ $12gh^7$	h $7a^3 b^3 c^2 \times b^5 c^6$ $7a^3 b^8 c^8$	i $x^4 \times y^3$ $x^4 y^3$
j $u^4 v^7 \times u^8 \times v^6$ $u^{12} v^{13}$	k $(x+3)^2 \times (x+3)^9$ $(x+3)^{11}$	l $(b+c) \times (b+c)^6$ $(b+c)^7$

7 Simplify the following using index laws.

a $x^3 y^3 \times x^4 y^2$ $x^7 y^5$	b $-x^7 y^3 \times x^2 y$ $-x^9 y^4$	c $5x^3 y^5 \times xy^4$ $5x^4 y^9$
d $xy^4 z \times 4xy$ $4x^2 y^5 z$	e $-3m^3 \times 5m^2$ $-15m^5$	f $4e^2 f^2 \times 2e^2 f^2$ $8e^4 f^4$
g $5c^4 d \times 4c^3 d$ $20c^7 d^2$	h $-9yz^2 \times -2yz^5$ $18y^2 z^7$	i $6m^{12} \times 3mn$ $18m^{13} n$
j $3x^4 \times 2x^3 \times 3x$ $18x^8$	k $-9x^5 \times 3x^2$ $-27x^7$	l $(-3m^3 n^2) \times (-4mn^7)$ $12m^4 n^9$

8 Simplify these expressions.

a $\frac{10m^7n^5}{2m^2n^4}$

$5m^5n$

b $\frac{12a^4b^5}{8ab^3}$

$\frac{3a^3b^2}{2}$

c $\frac{10h^8g^3}{25h^4g^3}$

$\frac{2h^4}{5}$

d $20x^2y^3 \div 2xy^5$

$\frac{10x}{y^2}$

9 Simplify:

a $3x^2y \times 4x^3y^4$

$12x^5y^5$

b $\frac{12m^7n^4}{4m^2n}$

$3m^5n^3$

c $2xy^3 \times 6x^4y$

$12x^5y^4$

d $5a^4b^2 \times 2ab^5$

$10a^5b^7$

e $\frac{20p^3q^5}{5pq^2}$

$4p^2q^3$

f $\frac{15a^9b^3}{3a^4b^2}$

$5a^5b$

10 Simplify the following using index laws.

a $\frac{5w^2x^5}{25wx^3}$

$\frac{wx^2}{5}$

b $\frac{7m^7y^4}{21m^2y^2}$

$\frac{m^5y^2}{3}$

c $\frac{6a^6b^{16}}{a^3b^7}$

$6a^3b^9$

d $\frac{16x^8y^6}{12x^2y^3}$

$\frac{4x^6y^3}{3}$

e $\frac{6s^6t^3}{14s^5t}$

$\frac{3st^2}{7}$

f $-\frac{5x^2y}{xy}$

$-5x$

11 Expand these brackets.

a $3x^5(2x^7 + 11)$

$6x^{12} + 33x^5$

b $4a^3(3b^{10} - 12a^8)$

$12a^3b^{10} - 48a^{11}$

c $6x^2y(5xy^5 + 3x^7 - y)$

$30x^3y^6 + 18x^9y - 6x^2y^2$

Mixed Practice: Basic Index Laws

FOUNDATION

1 Write each of the following in index form.

a $\frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3}$

$\left(\frac{2}{3}\right)^4$

b $\frac{3}{5} \times \frac{3}{5} \times \frac{3}{5} \times \frac{3}{5} \times \frac{3}{5}$

$\left(\frac{3}{5}\right)^5$

c $\frac{4}{7} \times \frac{4}{7} \times \frac{1}{5} \times \frac{1}{5} \times \frac{1}{5} \times \frac{1}{5}$

$\left(\frac{4}{7}\right)^2 \left(\frac{1}{5}\right)^4$

d $\frac{7x}{9} \times \frac{7x}{9} \times \frac{y}{4} \times \frac{y}{4} \times \frac{y}{4}$

$\left(\frac{7x}{9}\right)^2 \left(\frac{y}{4}\right)^3$

2 Find the missing numbers.

a $6^5 \times 6^7 = 6^{\square} + \square = 6^{\square}$

6^{12}

b $a^{13} \times a^2 = a^{\square} + \square = a^{\square}$

a^{15}

c $5^{12} \div 5^4 = 5^{\square} - \square = 5^{\square}$

5^8

d $\frac{m^{16}}{m^2} = m^{\square} - \square = m^{\square}$

m^{14}

3 Simplify

a $x^4 \times x^3$

x^7

b $b^6 \times b^4$

b^{10}

c $h \times h^7$

h^8

d $w^9 \div w^5$

w^4

e $\frac{a^7}{a^2}$

a^5

f $\frac{c^3}{c^9}$

$\frac{1}{c^6}$

g $\frac{m^{10}}{m^4}$

m^6

h $\frac{x^2}{x^8}$

$\frac{1}{x^6}$

i $(y^3)^2$

y^6

j $(p^5)^4$

p^{20}

k $(q^2)^6$

q^{12}

l $(t^4)^3$

t^{12}

m $5m^0$

5

n $(7n)^0$

1

o $-3r^0$

-3

p $(2a)^0 + 1$

2

q $4x^2 \times 3x^5$

$12x^7$

r $\frac{12a^8}{4a^3}$

$3a^5$

s $5x^3y^2 \times 2xy^4$

$10x^4y^6$

t $\frac{18m^7}{6m^3}$

$3m^4$

4

a Simplify
 $4y^3 \times 2y^8$

$$8y^{11}$$

b Work out the value of
 $(4 \times 10^3) \times (2 \times 10^8)$

$$8 \times 10^{11}$$

c Simplify
 $\frac{4y^3}{2y^8}$

$$\frac{2}{y^5}$$

d Work out the value of
 $\frac{4 \times 10^3}{2 \times 10^8}$

$$\frac{2}{10^5}$$

5 Simplify these expressions using the index laws.

a $x^4 \times x^3$

$$x^7$$

b $a^6 \times a^3$

$$a^9$$

c $t^5 \times t^3$

$$t^8$$

d $a^6 \div a^4$

$$a^2$$

e $x^5 \div x^2$

$$x^3$$

f $\frac{q^{12}}{q^2}$

$$q^{10}$$

g $(x^3)^5$

$$x^{15}$$

h $(x^7)^3$

$$x^{21}$$

i x^0

$$1$$

j $y \times y^4$

$$y^5$$

k $d^2 \times d$

$$d^3$$

l $\frac{d^7}{d^6}$

$$d$$

m $(y^7)^0$

$$1$$

n $2x^0$

$$2$$

o $\frac{e^5}{e^0}$

$$e^5$$

p $y^2 \times y \times y^4$

$$y^7$$

q $b \times b^5 \times b^2$

$$b^8$$

r $\frac{8b^{10}}{4b^5}$

$$2b^5$$

s $\frac{12d^{10}}{36d^5}$

$$\frac{d^5}{3}$$

t $q^6 \times q^3 \times q^2$

$$q^{11}$$

u $\frac{4a^{14}}{2a^7}$

$$2a^7$$

v $\frac{18y^{15}}{9y^7}$

$$2y^8$$

w $\frac{9y^7}{18y^{15}}$

$$\frac{1}{2y^8}$$

x $9m^3 \div m^2$

$$9m$$

y $5y^4 \div y^2$

$$5y^2$$

z $\frac{3m^7}{12m^2}$

$$\frac{m^5}{4}$$

aa $\frac{4a^4}{20a^3}$

$$\frac{a}{5}$$

6 Simplify the following.

a $(y^6)^2$	b $(m^3)^6$	c $(x^2)^5$
y^{12}	m^{18}	x^{10}
d 5^0	e $(-6)^0$	f $5(m^8)^2$
1	1	$5m^{16}$
g $4(q^7)^4$	h $(-3)^0$	i $-(4)^0$
$4q^{28}$	1	-1
j $-3(c^2)^5$	k $2(j^4)^6$	l $-3p^0$
$-3c^{10}$	$2j^{24}$	-3
m $(0.75)^0$	n $\left(\frac{3}{4}\right)^0$	o $(4y)^0$
1	1	1
p $5m^0$	q $-3p^0$	r $6x^0 - 2x^0$
5	-3	4
s $-5n^0 - (8n)^0$	t $(3x^4)^0$	u $1^0 + 2^0 + 3^0$
-6	1	3
v $(1 + 2 + 3)^0$	w $100^0 - a^0$	x $(-5)^0$
1	0	1

7 Simplify

a $\frac{6x^4y^3}{2xy}$	b $-2x^3 \times 5x^2$	c $4ab^3 \times 3a^4b$	d $(c^4)^2 \times (c^3)^2$
$3x^3y^2$	$-10x^5$	$12a^5b^4$	c^{14}
e $3a^2b \times 4a^3b^2$	f $\frac{a^7b^3}{a^2b^5}$	g $\frac{15x^6y^2}{3x^2y^4}$	h $\frac{m^9n^4}{m^4n^9}$
$12a^5b^3$	$\frac{a^5}{b^2}$	$\frac{5x^4}{y^2}$	$\frac{m^5}{n^5}$
i $(m^2)^4 \times m^3$	j $(y^3)^5 \div y^{10}$	k $h^4 \times (h^2)^3$	l $9x^0y \times 2y^3$
m^{11}	y^5	h^{10}	$18y^4$
m $\frac{20p^9}{5p^4}$	n $7m^0 + 3n^0$	o $\frac{-6p^5}{2p^2}$	
$4p^5$	10	$-3p^3$	

8 Simplify these expressions.

a $\frac{x^3 \times x^6}{x^4}$	b $\frac{y^3 \times y^2}{y^2}$	c $\frac{4^5 \times 4^3}{4^2}$
x^5	y^3	4^6
d $\frac{a^5 \times a^2}{a}$	e $\frac{p^2 \times p^3}{p^7}$	f $\frac{t^2}{t^4 \times t^2}$
a^6	$\frac{1}{p^2}$	$\frac{1}{t^4}$
g $\frac{3^4 \times 3^2 \times 3^2}{3^4}$	h $\frac{y^4 \times y^3}{y^2 \times y}$	i $\frac{2^2 \times 2^3 \times 2^2}{2^7 \times 2^3}$
3^4	y^4	$\frac{1}{2^3}$

9 Simplify

a $5 \times (5^5)^5$	b $(m^2)^5 \times m^4$	c $(x^3 \times x^4)^2$
5^{26}	m^{14}	x^{14}
d $(n^3)^7 \div n^5$	e $(m^3)^5 \times (m^4)^2$	f $\frac{g^{10}}{(g^2)^3}$
n^{16}	m^{23}	g^4

10

a Simplify $\frac{9y^4}{3y^7}$	b Simplify $9y^4 \times 3y^7$
$\frac{3}{y^3}$	$27y^{11}$
e Work out the value of $(9 \times 10^4) \times (3 \times 10^7)$	c Simplify $\frac{9 \times 10^4}{3 \times 10^7}$
27×10^{11}	$\frac{3}{10^3}$

11 Simplify these expressions.

a $(-x)^0$	b $-x^0$	c $w^0 - x^0$	d $s^0 - (-x)^0$
1	-1	0	0
e $w^0 + (-x)^0$	f $(-w)^0 + (-x)^0$	g $-w^0 + (-x)^0$	h $-w^0 - x^0$
2	2	0	-2

12 Find the missing numbers.

a $2^7 \times 2^{\square} = 2^{19}$	b $6^{\square} \times 6^3 = 6^{11}$	c $11^6 \div 11^{\square} = 11^3$
12	8	3
d $19^{\square} \div 19^2 = 19$	e $x^6 \times x^{\square} = x^7$	f $a^{\square} \times a^2 = a^{20}$
3	1	18
g $b^{13} \div b^{\square} = b$	h $y^{\square} \div y^9 = y^2$	i $\square \times x^2 \times 3x^4 = 12x^6$
12	11	4
j $15y^4 \div (\square y^3) = y$	k $\square a^9 \div (4a) = \frac{a^8}{2}$	l $13b^6 \div (\square b^5) = \frac{b}{3}$
15	2	39

13 Match the cards.

a $g^3(g^2 + g^5)$	A g^{10}
b $g^3(g^2 \times g^5)$	B g^{21}
c $(g^2)^3 + (g^5)^3$	C $g^5 + g^8$
d $(g^2)^3 \times (g^5)^3$	D $g^6 + g^{15}$

a - C, b - A, c - D, d - B

14 Simplify each of the following by combining various index laws

a $4 \times (4^3)^2$	b $(3^4)^2 \times 3$	c $x \times (x^0)^5$
4^7	3^9	x
d $7^8 \div (7^3)^2$	e $(4^2)^3 \div 4^5$	f $(3^6)^3 \div (3^5)^2$
7^2	4	3^8
g $y^5 \times (y^2)^4$	h $b^5 \times (b^3)^3$	i $(a^2)^3 \times a^4$
y^{13}	b^{14}	a^{10}
j $(m^3)^6 \div (m^2)^9$	k $(y^5)^3 \div (y^6)^2$	l $(h^{11})^2 \div (h^5)^4$
1	y^3	h^2
m $a^3f \times (a^4)^2 \times (f^4)^3$	n $x^2y \times (x^3)^4 \times (y^2)^2$	o $(s^2)^3 \times 5(r^0)^3 \times rs^2$
$a^{11}f^{13}$	$x^{14}y^5$	$5rs^8$

15 Simplify these expressions.

a $\frac{a^3 \times a^5}{a^6}$	f $2x^3 \times 3x^5$
a^2	$6x^8$
b $\frac{a^6}{a^3 \times a^5}$	g $2x^3y^2 \times 3x^5y^2$
$\frac{1}{a^2}$	$6x^8y^4$
c $\frac{x^6}{a^3 \times a^5}$	h $12x^6y^2 \times 3x^5y^2$
$\frac{x^6}{a^8}$	$36x^{11}y^4$
d $\frac{12x^6}{2a^3 \times 3a^5}$	i $12x^6y^2 \div 3x^5y^2$
$\frac{2x^6}{a^8}$	$4x$
e $\frac{12x^6}{2x^3 \times 3x^5}$	j $\frac{12x^6y^8}{3x^5y^2}$
$\frac{2}{x^2}$	$4xy^6$

16 Look at the example and then answer the questions.

$$\frac{v^4x^5 \times 5v^3}{4vx^3 \times 3xv} = \frac{5v^7x^5}{12v^2x^4} = \frac{5v^5x}{12}$$

a $\frac{3h^2c \times 3h^4}{2h^3 \times c^3} = \frac{\boxed{}}{\boxed{}} = \frac{\boxed{}}{\boxed{}} = \frac{9h^3}{2c^2}$	b $\frac{5a^2y^2 \times 2y^4}{7y^4 \times a^3} = \frac{\boxed{}}{\boxed{}} = \frac{\boxed{}}{\boxed{}} = \frac{10y^2}{7a}$	c $\frac{2j^2k \times 4k^2}{3j^4} = \frac{\boxed{}}{\boxed{}} = \frac{\boxed{}}{\boxed{}} = \frac{8k^3}{3j^2}$
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17 Read the example carefully and answer the questions.

Liz's work

Write as an expression without brackets:

$b^3(b^2)^4$

$b^3(b^2)^4$

$b^3 \cdot b^{2 \cdot 4}$

$b^3 \cdot b^8$

b^{3+8}

b^{11}

In the step marked with an arrow, why did Liz add instead of multiply?

Simplify

Index law of multiplication says to add

$$(r^5)^2r^3$$

$$r^{13}$$

18 These simplifications are not correct. Provide the correct answer.

a $a^4 \times a = a^4$ ✗ a^5	b $x^7 \div x = x^7$ ✗ x^6	c $3a^5 \div 6a^3 = 2a^2$ ✗ $\frac{a^2}{2}$
d $5x^7 \div 10x^3 = \frac{1}{2x^4}$ ✗ $\frac{x^4}{2}$	e $2x^7 \times 3x^4 = 5x^{11}$ ✗ $6x^{11}$	f $a^5 \div a^2 \times a = a^5 \div a^3$ ✗ a^4
g $(a^4)^5 = a^9$ ✗ a^{20}	h $3(x^3)^2 = 9x^6$ ✗ $3x^6$	i $(-2x)^0 = -2$ ✗ 1

19

a Simplify

$$\frac{(a^3)^2 \times a^4}{(a^2)^5}$$

1

b Does the value of a matter?

no

20 NAPLAN B

Simplify the expression $\frac{(12x)^0}{4} \times \frac{5x}{3}$

$\frac{5x}{12}$

21 NAPLAN B

Simplify the expression $\frac{20m^0}{n} \times \frac{n^2}{5m}$

$\frac{4n}{m}$

22 Simplify these expressions.

a

$$\frac{a^5 \times a^4}{a^6} = \frac{\boxed{}}{\boxed{}} = \underline{\underline{\boxed{}}}$$

a^3

b

$$\frac{n^3}{n^7 \times n^2} = \frac{\boxed{}}{\boxed{}} = \underline{\underline{\boxed{}}}$$

$\frac{1}{n^6}$

c

$$\frac{g^5}{g^2 \times g^4} = \frac{\boxed{}}{\boxed{}} = \underline{\underline{\boxed{}}}$$

$\frac{1}{g}$

d

$$\frac{p^5}{p^3 \times p^4} = \frac{\boxed{}}{\boxed{}} = \underline{\underline{\boxed{}}}$$

$\frac{1}{p^2}$

e

$$\frac{d^3 \times d^5}{d \times d^4} = \frac{\boxed{}}{\boxed{}} = \underline{\underline{\boxed{}}}$$

d^3

f

$$\frac{c^5 \times c^4}{c^3 \times c^2} = \frac{\boxed{}}{\boxed{}} = \underline{\underline{\boxed{}}}$$

c^4

g

$$\frac{ab \times b^2}{a^2 \times a^3}$$

$\frac{b^3}{a^4}$

h

$$\frac{mx^5 \times m^2}{m^4 \times xm}$$

$\frac{x^4}{m^2}$

i

$$\frac{xy^5 \times y^2}{x^4 \times x^2y}$$

$\frac{y^6}{x^5}$

j

$$\frac{cn^3 \times cn^2}{c^2 \times cn^2}$$

$\frac{n^3}{c}$

k

$$\frac{ac^3 \times a^2}{a^3c^4 \times c^2}$$

$\frac{1}{c^3}$

l

$$\frac{w^2n^5 \times w^2}{nw \times w^3}$$

n^4

m

$$\frac{k^2y \times y^2k^4}{y^3 \times y^2k^4}$$

$\frac{k^2}{y^2}$

n

$$\frac{q^3n^2 \times n^4}{qn^6 \times n^4q^3}$$

$\frac{1}{qn^4}$

o

$$\frac{b^2d^3 \times b^2d}{bd^2 \times b^2d^2}$$

b

23 Look at the example and then answer the questions.

$$\frac{3v^4x^5 \times v^3}{2vx^3 \times 6xv} = \frac{3v^7x^5}{12v^2x^4} = \frac{v^5x}{4}$$

a $\frac{3ve^3 \times 5e}{2e^3v^5}$

b $\frac{2n^2w \times 6n^4}{3n^3}$

c $\frac{2x^2y^3 \times 5y^4}{10x \times 2y^3}$

$$\frac{15e}{2v^4}$$

$$4n^3w$$

$$\frac{xy^4}{2}$$

d $\frac{5m^2e^3 \times m^2}{2m^4 \times e^4m}$

e $\frac{6h^2q \times 3q^2}{2hq^4 \times h^2q^2}$

f $\frac{5y^2x^5 \times 4y^2}{y^4x^3 \times 10x^2}$

$$\frac{5}{2me}$$

$$\frac{9}{hq^3}$$

$$2$$

24 Simplify each of the following by combining various index laws

a $\frac{(b^2)^5}{b^4}$

b $\frac{(x^4)^3}{x^7}$

c $\frac{(y^3)^3}{y^3}$

$$b^6$$

$$x^5$$

$$y^6$$

d $\frac{3x^4 \times 6x^3}{9x^{12}}$

e $\frac{5x^5 \times 4x^2}{2x^{10}}$

f $\frac{24(x^4)^4}{8(x^4)^2}$

$$\frac{2}{x^5}$$

$$\frac{10}{x^3}$$

$$3x^8$$

g $\frac{4(d^4)^3 \times (e^4)^2}{8(d^2)^5 \times e^7}$

h $\frac{6(m^3)^2(n^5)^3}{15(m^5)^0(n^2)^7}$

i $\frac{2(a^3)^4(b^2)^6}{16(a^0)(b^6)^2}$

$$\frac{d^2e}{2}$$

$$\frac{2m^6n}{5}$$

$$\frac{a^{12}}{8}$$

25 Simplify the following using index laws.

a $b^5 \times b^2 \div b$	b $y^5 \times y^4 \div y^3$	c $c^4 \div c \times c^4$
b^6	y^6	c^7
d $x^4 \times x^2 \times x^5$	e $\frac{t^4 \times t^3}{t^6}$	f $\frac{p^2 \times p^7}{p^3}$
x^{11}	t	p^6
g $\frac{d^5 \times d^3}{d^2}$	h $\frac{x^9 \times x^2}{x}$	i $\frac{3x^3y^4 \times 8xy}{6x^2y^2}$
d^6	x^{10}	$4x^2y^3$
j $\frac{9b^4}{2g^3} \times \frac{4g^4}{3b^2}$	k $\frac{24m^7n^5}{5m^3n} \times \frac{5m^2n^4}{8mn^2}$	l $\frac{p^4q^3}{p^2q} \times \frac{p^6q^4}{p^3q^2}$
$6b^2g$	$3m^5n^6$	p^5q^4

26 Simplify $\frac{5ab^2}{2ab^2} \div \frac{10a^4b^7}{4a^3b^8}$

$\frac{b}{a}$

27



- a** Rhys picks two cards and then divides the expressions. He gets the answer y^3 . Which two cards does he pick? Find two different answers.

$10y^4 \div 10y,$ $12y^3 \div 12$

- b** Huda picks two cards and then divides the expressions. She gets the answer $\frac{3}{y^2}$. Which two cards does she pick?

$18y^6 \div 6y^8$

28 Find the missing terms.

a $\frac{2c^3}{\boxed{}} = 2c$ c^2	b $\frac{w^4}{\boxed{}} = \frac{w^3}{m}$ wm	c $\frac{\boxed{}}{3a} = 4$ $12a$	d $\frac{4e^5}{\boxed{}} = \frac{e^3}{2}$ $8e^2$
e $\boxed{} \div 2k = 3k$ $6k^2$	f $6a^2b \div \boxed{} = 3ab$ $2a$	g $\boxed{} \div y = \frac{3x}{y}$ $3x$	h $\frac{\boxed{}}{2j^3} = \frac{7}{j}$ $14j^2$
i $\frac{5ux}{\boxed{}} = \frac{1}{2w}$ $10uxw$	j $\boxed{} \div 4v^2y = \frac{-5y^3}{2v}$ $-20vy^4$	k $\frac{\boxed{}}{-2hg^3} = \frac{h}{g}$ $-2h^2g^2$	l $16x^3k \div \boxed{} = \frac{-8}{3x}$ $-6x^4k$
m $\boxed{} \div 2y^2 = 9x$ $18xy^2$	n $\frac{-8ka^3}{\boxed{}} = \frac{2k}{a^3}$ $-4a^6$	o $\boxed{} \div 4ey^4 = \frac{-1}{y^2}$ $-4ey^2$	

29 Given that $\frac{y^n \times y^5}{y^6} = y^{13}$, work out the value of n .

14

30 Find the value of n

$$\frac{(x^{-6})^5}{x^4} = x^n$$

-34

31 Evaluate without a calculator.

a $(2^4)^8 \div 2^{30}$ 4	b $(10^3)^7 \div 10^{18}$ 1000	c $(7^4)^9 \div 7^{36}$ 1
d $((-1)^{11})^2 \div ((-1)^2)^{11}$ 1	e $-2((-2)^3)^3 \div (-2)^8$ 4	f $\frac{(5^2)^3}{(8^4)^7} \times \frac{(8^7)^4}{(5^3)^2}$ 1

32 Split these expressions according to prime factors.

a 6^n

$$= (2 \times 3)^n$$

$$= 2^n \times 3^n$$

b 21^y

$$3^y \times 7^y$$

c 12^x

$$2^{2x} \times 3^x$$

d 10^{x+y}

$$2^{x+y} \times 5^{x+y}$$

e 35^{2x+3}

$$5^{2x+3} \times 7^{2x+3}$$

f 26^{4x-3z}

$$2^{4x-3z} \times 13^{4x-3z}$$

33 Simplify the following.

a $\frac{3a^5bc^3}{6a^4c} \times \frac{4b^3}{2abc} \times \frac{2a^3b^2c}{2a^2b^4c^2}$

$$ab$$

b $\frac{3a^4b^5}{a^5b^2} \div \frac{6b^3}{2a^2b}$

$$ab$$

c $\frac{2a}{3a^2b^3} \times \frac{9a^4b^7}{ab^5} \div \frac{6a}{b^2}$

$$ab$$

34 Expand:

a $6a^2(2a - 4a^3)$

$$12a^3 - 24a^5$$

b $c^{\frac{3}{2}} \left(c^{\frac{1}{2}} + c^{-\frac{3}{2}} \right)$

$$c^2 + 1$$

c $-6a^2b^3c(2a^3b + a^2bc^{-1})$

$$-12a^5b^4c - 6a^4b^4$$

35 Simplify

$$\frac{3x^{-\frac{5}{4}} \times 4x^{\frac{7}{4}}}{x^{\frac{3}{2}}}$$

$$\frac{12}{x} \text{ or } 12x^{-1}$$

36 Solve $3^x \times 3^{x+3} = 81$

$$x = 0.5$$

37 Solve to find x

$$\left(m^{\frac{3}{11}}\right)^x = m^{\frac{5}{9}}$$

Power of a Product

Review

☐ Simplify expressions using index law of multiplication

a $3x \times 3x$

b $(3x)^2$

c $2x^5 \times 2x^5 \times 2x^5$

d $(2x^5)^3$

☐ Evaluate powers

a 2^3

b $(-2)^3$

c 5^3

d $(-5)^2$

e 3^2

f $(-3)^3$

Power of a Product

$$(ab)^m = a^m b^m$$

Simplify Power of a product

When there is a product inside brackets, we apply the power to each factor inside brackets.

$$\begin{aligned} (5b)^2 \\ = 5^2 b^2 \end{aligned}$$

$$\begin{aligned} 5b^2 \\ = 5b^2 \end{aligned}$$

$$\begin{aligned} (3x^2y)^4 \\ = 3^4 x^8 y^4 \end{aligned}$$

$$\begin{aligned} 2(xy^2)^3 \\ = 2x^3 y^6 \end{aligned}$$

$$\begin{aligned} 4xy^3 \\ = 4xy^3 \end{aligned}$$

Simplify the following expressions:

a $(2a)^3$

b $3a^2$

c $(3a^2)^4$

d $4(b^2c)^3$

e $(xy)^4$

f $5xy^4$

Example Simplify expressions using power of a product

Simplify

$$\begin{aligned}(5b)^3 \\ &= 5^3 b^3 \\ &= 125b^3\end{aligned}$$

$$\begin{aligned}(-3x^4y)^2 \\ &= (-3)^2 x^8 y^2 \\ &= 9x^8 y^2\end{aligned}$$

$$\begin{aligned}4x(c^2d^3)^5 \\ &= 4xc^{10}d^{15}\end{aligned}$$

Guided Practice

Simplify

a $(2x)^4$

$$16x^4$$

b $(3y^2)^3$

$$27y^6$$

c $(-4a)^2$

$$16a^2$$

d $(2a^2b)^3$

$$8a^6b^3$$

e $3(x^2)^4$

$$3x^8$$

f Compare $(3x)^2$ and $3x^2$

$$9x^2, 3x^2$$

Key Ideas

1. When raising a product to a power, the power applies to factor inside brackets.
2. $(2xy^2z)^3 = \dots\dots\dots$
3. $2x(y^2z)^3 = \dots\dots\dots$

1 Complete the following

a

$$\begin{aligned}(5a)^3 &= 5a \times \square \times \square \\ &= 5 \times 5 \times 5 \times a \times \square \times \square \\ &= 5^3 \times \square\end{aligned}$$

$$5^3 \times a^3$$

b

$$\begin{aligned}(ab)^4 &= ab \times \square \times \square \times \square \\ &= a \times \square \times \square \times \square \times b \times \square \times \square \times \square \\ &= a^4 \times \square\end{aligned}$$

$$a^4 \times b^4$$

2 Simplify these expressions by expanding first, then multiplying.

a

$$\begin{aligned}(2x^4)^3 &= 2x^4 \times 2x^4 \times 2x^4 \\ &= 2^3 x^{12} = \dots\dots\dots\end{aligned}$$

b

$$\begin{aligned}(3h^5)^2 &= \square \times \square = 3^\square h^\square \\ &= \square\end{aligned}$$

$$9h^{10}$$

c

$$\begin{aligned}(5a^7)^3 &= \square \times \square \times \square = 5^\square a^\square \\ &= \square\end{aligned}$$

$$125a^{21}$$

d

$$\begin{aligned}(2n^6)^4 &= \square \times \square \times \square \times \square = 2^\square n^\square \\ &= \square\end{aligned}$$

$$16n^{24}$$

3 Simplify

a $(2x)^3$

$$8x^3$$

b $(4x)^2$

$$16x^2$$

c $(6y)^2$

$$36y^2$$

d $(3b)^3$

$$27b^3$$

e $(3x)^2$

$$9x^2$$

f $(5x)^3$

$$125x^3$$

g $(2a)^4$

$$16a^4$$

h $(10c)^2$

$$100c^2$$

4 Simplify

a $(ab)^3$

$$a^3b^3$$

b $(mn)^5$

$$m^5n^5$$

c $(3xy)^2$

$$9x^2y^2$$

d $(xy^2)^3$

$$x^3y^6$$

e $(xy)^4$

$$x^4y^4$$

f $(2ab)^3$

$$8a^3b^3$$

g $(a^2b)^4$

$$a^8b^4$$

h $(2a^2b^3)^3$

$$8a^6b^9$$

5 Simplify

a $(x^2)^3$

$$x^6$$

b $(3a^2)^2$

$$9a^4$$

c $(5y^3)^2$

$$25y^6$$

d $(2q^3)^4$

$$16q^{12}$$

e $(2x^2)^3$

$$8x^6$$

f $(2m^4)^3$

$$8m^{12}$$

g $(4p^5)^2$

$$16p^{10}$$

h $(3r^2)^3$

$$27r^6$$

6 Simplify these expressions using the rule.

a $(2x^4)^3 = 2^3 x^{12} = \dots\dots\dots$	b $(3e^2)^3 = 3^{\square} e^{\square} = \square$ $27e^6$	c $(5n^4)^2 = 5^{\square} n^{\square} = \square$ $25n^8$
d $(2c^5)^4 = 2^{\square} c^{\square} = \square$ $16c^{20}$	e $(3x^3)^4 = 3^{\square} x^{\square} = \square$ $81x^{12}$	f $(3k^4)^3 = \square$ $27k^{12}$
g $(4c^5)^4 = \square$ $256c^{20}$	h $(6v^5)^3 = \square$ $216v^{15}$	i $(10a^8)^5 = \square$ $100\,000a^{40}$

7

a Complete the following:

$$(x^3y^2)^3 = x^3y^2 \times \square \times \square = \square$$

x^9y^6

b Simplify these expressions

i $(a^4b^3)^2$ a^8b^6	ii $(f^3g^6)^4$ $f^{12}g^{24}$	iii $(tm^6)^3$ t^3m^{18}
iv $(2p^5q^2)^4$ $16p^{20}q^8$	v $(3m^{10}n^6)^3$ $27m^{30}n^{18}$	vi $(a^3b^5c^7)^5$ $a^{15}b^{25}c^{35}$

8 Simplify.

a $(2x)^3$ $8x^3$	b $(5y)^2$ $25y^2$	c $(4a^2)^3$ $64a^6$	d $(-3r)^2$ $9r^2$
e $(a^3b)^5$ $a^{15}b^5$	f $(x^3y^2)^4$ $x^{12}y^8$	g $(c^{10}e^2)^8$ $c^{80}e^{16}$	h $(a^4b^2c)^5$ $a^{20}b^{10}c^5$
i $(3x^5)^2$ $9x^{10}$	j $(2x^3)^4$ $16x^{12}$	k $(5p^3)^3$ $125p^9$	l $\left(\frac{1}{2}g^4\right)^3$ $\frac{g^{12}}{8}$
m $-(3r)^2$ $-9r^2$	n $(-2h^2)^4$ $16h^8$	o $(5c^2d^3)^4$ $625c^8d^{12}$	p $(2x^3y^2)^5$ $32x^{15}y^{10}$

9 Read the example carefully and answer the questions.

$$(ab^5)^2$$

$$ab^{10}$$

By expanding and writing out $(ab^5)^2$, show why this is wrong.

Simplify $(cj^3)^5$

$$ab^5 \times ab^5 = a^2b^{10}$$

$$(-10a^6)^2 \cdot a^2$$

$$(-10a^6)^2 \cdot a^2$$

$$(-10)^2 \cdot (a^6)^2 \cdot a^2$$

$$100 \cdot a^{12} \cdot a^2$$

$$100 \cdot a^{14}$$

$$100a^{14}$$

Explain why we need brackets for $(-10)^2$.

Simplify

$$\text{Without brackets, } -10^2 = -100$$

$$(-2b^3)^2 \times b^4$$

10 Simplify.

a $(-2x)^2$

$$4x^2$$

b $(-3a)^2$

$$9a^2$$

c $(-5y^2)^2$

$$25y^4$$

d $(-2p^4)^3$

$$-8p^{12}$$

e $(-2x)^3$

$$-8x^3$$

f $(-3a)^3$

$$-27a^3$$

g $(-4m^3)^2$

$$16m^6$$

h $(-y)^5$

$$-y^5$$

11 Simplify

a $2(x^2)^3$

$$2x^6$$

b $5(m^3)^2$

$$5m^6$$

c $2(pq)^4$

$$2p^4q^4$$

d $-2(x^3)^4$

$$-2x^{12}$$

e $3(a^4)^2$

$$3a^8$$

f $4(y^5)^3$

$$4y^{15}$$

g $3(a^2b)^3$

$$3a^6b^3$$

h $5(2x)^3$

$$40x^3$$

12 Find the area of these squares.

a $(3a)^2 = \dots 3a \dots$

$$9a^2$$

b $4b^4$

c a^4b^6

d $9x^4y^2$

13 Find the missing numbers.

$$(10y^{\boxed{}})^{\boxed{}} = 100\,000y^{30}$$

$$(10y^6)^5$$

14 Simplify these pairs of expressions.

a $4x^2$ and $(4x)^2$

$$4x^2, \quad 16x^2$$

b $-6y^2$ and $(-6y)^2$

$$-6y^2, \quad 36y^2$$

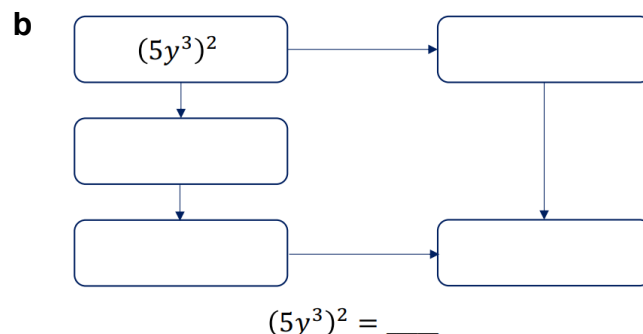
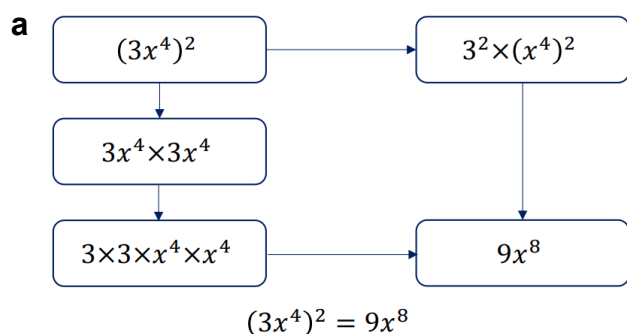
c $-2a^3$ and $(-2a)^3$

$$-2a^3, \quad -8a^3$$

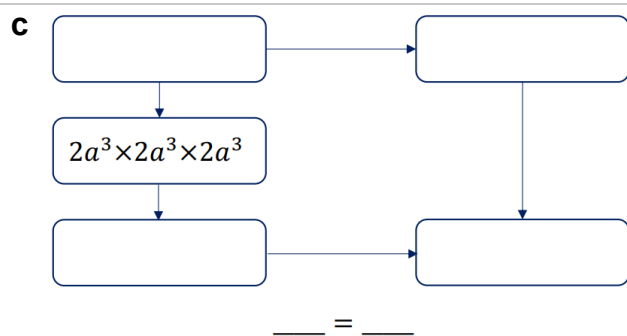
d $-3(x^2y)^3$ and $x^2(-3y)^3$

$$-3x^6y^3, \quad -27x^2y^3$$

15 Use two methods to simplify these expressions. The first is done for you.



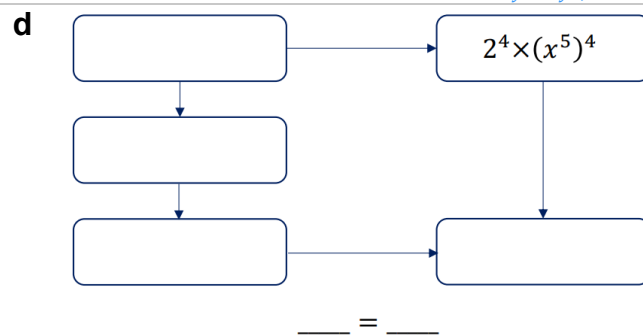
Top to bottom:
 $5^2 \times (y^3)^2$
 $5y^3 \times 5y^3$
 $5 \times 5 \times y^3 \times y^3, \quad 25y^6$



$$(2a^3)^3, \quad 2^3 \times (a^3)^3$$

$$2 \times 2 \times 2 \times a^3 \times a^3 \times a^3, \quad 8a^9$$

$$(2a^3)^3 = 8a^9$$



$$(2x^5)^4$$

$$2x^5 \times 2x^5 \times 2x^5 \times 2x^5$$

$$2 \times 2 \times 2 \times 2 \times x^5 \times x^5 \times x^5 \times x^5, \quad 16x^{20}$$

$$(2x^5)^4 = 16x^{20}$$

16 Expand then simplify these expressions.

a $(x^2b)^3 \times (x^4b^2)^2$

$$= \underline{x^6b^3 \times x^8b^4} \quad \text{Expand}$$

$$= \underline{x^{14}b^7} \quad \text{Simplify}$$

b $(x^3y^2)^4 \times (xy^5)^2$

$$x^{14}y^{18}$$

c $(b^2k)^3 \times (b^2k^2)^4$

$$b^{14}k^{11}$$

d $(u^6v^3)^2 \times (uv^2)^3$

$$u^{15}v^{12}$$

e $(a^2h^3)^5 \times (2h^2)^2$

$$4a^{10}h^{19}$$

Power of a Quotient

Review

☐ Simplify expressions using index law of multiplication

a $\frac{x}{3} \times \frac{x}{3}$

b $\left(\frac{x}{3}\right)^2$

c $\frac{2}{x^5} \times \frac{2}{x^5} \times \frac{2}{x^5}$

d $\left(\frac{2}{x^5}\right)^3$

☐ Evaluate powers

a $\left(\frac{1}{2}\right)^3$

b $\left(\frac{2}{3}\right)^3$

c $\left(\frac{3}{5}\right)^2$

i Power of a Quotient

$$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$$

? Simplify Power of a quotient

For fractions inside brackets, we apply the power to each term in numerator and denominator.

$$\left(\frac{x}{y}\right)^3 = \frac{x^3}{y^3}$$

$$\left(\frac{ab^2}{c^4}\right)^3 = \frac{a^3b^6}{c^{12}}$$

$$x\left(\frac{a^2}{b^5}\right)^4 = \frac{xa^8}{b^{20}}$$

Simplify the following expressions:

a $\left(\frac{a}{b}\right)^4$

b $\left(\frac{x^2}{y}\right)^5$

c $\left(\frac{pq^2}{r^{11}}\right)^9$

d $2\left(\frac{a}{b}\right)^4$

e $3x\left(\frac{a}{b}\right)^5$

f $2x\left(\frac{a^2}{b^7}\right)^3$

Example Simplify expressions using power of a product

Simplify

$$\left(\frac{6}{b}\right)^3 = \frac{6^3}{b^3}$$

$$= \frac{216}{b^3}$$

$$\left(\frac{ab^2}{c^4}\right)^4 = \frac{a^4b^8}{c^{16}}$$

$$\left(\frac{-2}{3b}\right)^4 = \frac{(-2)^4}{3^4b^4}$$

$$= \frac{16}{81b^4}$$

Guided Practice

Simplify

a $\left(\frac{p}{q}\right)^2$

$$\frac{p^2}{q^2}$$

b $\left(\frac{4}{y}\right)^3$

$$\frac{64}{y^3}$$

c $\left(\frac{-5}{p^2}\right)^4$

$$\frac{625}{p^8}$$

d $\left(\frac{2x}{y}\right)^5$

$$\frac{32x^5}{y^5}$$

e $3\left(\frac{n^3}{2m}\right)^2$

$$\frac{3n^6}{4m^2}$$

f $\left(\frac{-3f}{2g^2}\right)^3$

$$\frac{-27f^3}{8g^6}$$

Key Ideas

- When raising a fraction to a power, the power applies to both the and

1 Complete the following

$$\begin{aligned} \mathbf{a} \quad \left(\frac{x}{6}\right)^3 &= \frac{x}{6} \times \square \times \square \\ &= \frac{x \times \square \times \square}{6 \times \square \times \square} \\ &= \frac{x^3}{\square} \end{aligned}$$

$$\frac{x^3}{6^3}$$

$$\begin{aligned} \mathbf{b} \quad \left(\frac{a}{b}\right)^5 &= \frac{a}{b} \times \square \times \square \times \square \times \square \\ &= \frac{a \times \square \times \square \times \square \times \square}{b \times \square \times \square \times \square \times \square} \\ &= \frac{a^5}{\square} \end{aligned}$$

$$\frac{a^5}{b^5}$$

2 Simplify

$$\mathbf{a} \quad \left(\frac{a}{b}\right)^3$$

$$\frac{a^3}{b^3}$$

$$\mathbf{b} \quad \left(\frac{m}{n}\right)^2$$

$$\frac{m^2}{n^2}$$

$$\mathbf{c} \quad \left(\frac{c}{d}\right)^6$$

$$\frac{c^6}{d^6}$$

$$\mathbf{d} \quad \left(\frac{x}{y}\right)^4$$

$$\frac{x^4}{y^4}$$

$$\mathbf{e} \quad \left(\frac{p}{q}\right)^5$$

$$\frac{p^5}{q^5}$$

$$\mathbf{f} \quad \left(\frac{r}{s}\right)^3$$

$$\frac{r^3}{s^3}$$

3 Simplify

$$\mathbf{a} \quad \left(\frac{2}{x}\right)^3$$

$$\frac{8}{x^3}$$

$$\mathbf{b} \quad \left(\frac{4}{y}\right)^2$$

$$\frac{16}{y^2}$$

$$\mathbf{c} \quad \left(\frac{x}{2}\right)^4$$

$$\frac{x^4}{16}$$

$$\mathbf{d} \quad \left(\frac{3}{a}\right)^2$$

$$\frac{9}{a^2}$$

$$\mathbf{e} \quad \left(\frac{5}{m}\right)^3$$

$$\frac{125}{m^3}$$

$$\mathbf{f} \quad \left(\frac{a}{3}\right)^3$$

$$\frac{a^3}{27}$$

4 Simplify

$$\mathbf{a} \quad \left(\frac{p}{q}\right)^3$$

$$\frac{p^3}{q^3}$$

$$\mathbf{b} \quad \left(\frac{x}{y}\right)^4$$

$$\frac{x^4}{y^4}$$

$$\mathbf{c} \quad \left(\frac{4}{y}\right)^3$$

$$\frac{64}{y^3}$$

$$\mathbf{d} \quad \left(\frac{5}{p^2}\right)^4$$

$$\frac{625}{p^8}$$

$$\mathbf{e} \quad \left(\frac{2}{r^3}\right)^2$$

$$\frac{4}{r^6}$$

$$\mathbf{f} \quad \left(\frac{s^3}{7}\right)^2$$

$$\frac{s^6}{49}$$

$$\mathbf{g} \quad \left(\frac{2m}{n}\right)^5$$

$$\frac{32m^5}{n^5}$$

$$\mathbf{h} \quad \left(\frac{2a^2}{3}\right)^3$$

$$\frac{8a^6}{27}$$

$$\mathbf{i} \quad \left(\frac{3n^3}{2m^4}\right)^3$$

$$\frac{27n^9}{8m^{12}}$$

5 Simplify

a $\left(\frac{x^2}{y}\right)^3$

$\frac{x^6}{y^3}$

b $\left(\frac{m^4}{n^2}\right)^3$

$\frac{m^{12}}{n^6}$

c $\left(\frac{x^3}{y^3}\right)^2$

$\frac{x^6}{y^6}$

d $\left(\frac{a}{b^3}\right)^2$

$\frac{a^2}{b^6}$

e $\left(\frac{p^2}{q^5}\right)^4$

$\frac{p^8}{q^{20}}$

f $\left(\frac{c^2d}{e^3}\right)^2$

$\frac{c^4d^2}{e^6}$

6 Simplify

a $\left(\frac{3x}{2y}\right)^2$

$\frac{9x^2}{4y^2}$

b $\left(\frac{4m}{n}\right)^2$

$\frac{16m^2}{n^2}$

c $\left(\frac{2x^2}{3y}\right)^2$

$\frac{4x^4}{9y^2}$

d $\left(\frac{2a}{5b}\right)^3$

$\frac{8a^3}{125b^3}$

e $\left(\frac{p}{3q}\right)^3$

$\frac{p^3}{27q^3}$

f $\left(\frac{5a}{2b^3}\right)^2$

$\frac{25a^2}{4b^6}$

7 Simplify

a $\left(\frac{-2}{x}\right)^2$

$\frac{4}{x^2}$

b $\left(\frac{-3a}{b}\right)^2$

$\frac{9a^2}{b^2}$

c $\left(\frac{-1}{2m}\right)^3$

$-\frac{1}{8m^3}$

d $\left(\frac{-2}{x}\right)^3$

$-\frac{8}{x^3}$

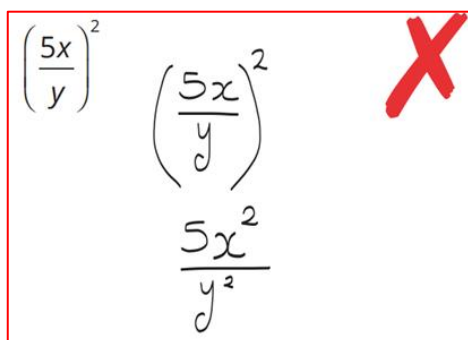
e $\left(\frac{-x}{y}\right)^4$

$\frac{x^4}{y^4}$

f $\left(\frac{-3}{2a^2}\right)^2$

$\frac{9}{4a^4}$

8 Read the example carefully and answer the questions.



$$\left(\frac{5x}{y}\right)^2 \quad \left(\frac{5x}{y}\right)^2 \quad \frac{5x^2}{y^2}$$

There is something wrong with the 5 in the answer. What should there be instead?

25

Write as an expression without brackets:

$\left(\frac{3a}{b}\right)^4$

$\frac{81a^4}{b^4}$

9 Simplify these fractions without a calculator. The first one is done for you.

a	$\frac{2^4}{6^4} = \left(\frac{2}{6}\right)^4$ $= \left(\frac{1}{3}\right)^4$ $= \frac{1}{81}$	b	$\frac{6^3}{3^3}$	c	$\frac{10^4}{5^4}$	d	$\frac{3^3}{30^3}$
			8		16		$\frac{1}{1000}$

10 Solve these equations.

a	$\left(\frac{a}{3}\right)^2 = \frac{25}{9}$	b	$\left(\frac{a}{2}\right)^4 = 16$	c	$(5a)^3 = 1000$
	$a = \pm 5$		$a = \pm 4$		$a = 2$
d	$(2a)^4 = 256$	e	$\left(\frac{2a}{5}\right)^2 = \frac{36}{25}$	f	$\left(\frac{6a}{7}\right)^3 = 1728$
	$a = \pm 2$		$a = \pm 3$		$a = \sqrt[3]{2744} = 14$

11 The rule for the number of seeds germinating in a glass house over a two-week period is given by $N = \left(\frac{t}{2}\right)^3$, where N is the number of germinating seeds and t is the number of days.

- a** Find the number of germinating seeds after:
- i** 4 days
 8
 - ii** 10 days
 125
- b** Rewrite the rule without brackets using index laws
 $N = \frac{t^3}{8}$
- c** Use your rule in part **b** to find the number of days required to germinate:
- i** 64 seeds
 $t = \sqrt[3]{512} = 8$
 - ii** 1 seed
 $t = \sqrt[3]{8} = 2$

Mixed Practice: Power of a Product and Quotient

FOUNDATION

1 Simplify:

a $(3x)^2$ $9x^2$	b $\left(\frac{x}{2}\right)^3$ $\frac{x^3}{8}$	c $\left(\frac{2m}{n}\right)^2$ $\frac{4m^2}{n^2}$	d $\frac{15p^8}{5p^3}$ $3p^5$
e $\left(\frac{a}{b}\right)^3$ $\frac{a^3}{b^3}$	f $(p^4)^3$ p^{12}	g $h^5 \times h^4$ h^9	h $(xy^2)^3$ x^3y^6
i $5m^4 \times 2m^3$ $10m^7$	j $(4b)^0$ 1	k $(c^3d)^2$ c^6d^2	l $\left(\frac{3}{x}\right)^2$ $\frac{9}{x^2}$
m $(2y^3)^2$ $4y^6$	n $\frac{x^3}{x^9}$ $\frac{1}{x^6}$	o $7x^0 + 2$ 9	p $3a^2b \times 4ab^5$ $12a^3b^6$
q $\frac{12a^6}{4a^2}$ $3a^4$	r $(-2a)^3$ $-8a^3$	s $\left(\frac{a^2}{b}\right)^4$ $\frac{a^8}{b^4}$	t $(-3m^2)^2$ $9m^4$

2 Simplify:

a $(2x^2y)^3$ $8x^6y^3$	b $\left(\frac{x^3}{y^2}\right)^3$ $\frac{x^9}{y^6}$	c $\left(\frac{2x^2}{3y}\right)^3$ $\frac{8x^6}{27y^3}$	d $(2a^2)^3 \times (3a)^2$ $72a^8$
e $\left(\frac{3a}{2b}\right)^2$ $\frac{9a^2}{4b^2}$	f $-4(a^2)^3$ $-4a^6$	g $\frac{(3m^2n)^2}{3mn}$ $3m^{3n}$	h $\left(\frac{-3m}{n^2}\right)^2$ $\frac{9m^2}{n^4}$
i $\frac{(2x)^3}{x}$ $8x^2$	j $\frac{(5p^2)^2}{5p}$ $5p^3$	k $(-2x)^4 \times x^3$ $16x^7$	l $\frac{(4x^2y)^2}{2xy}$ $8x^{3y}$
m $(3m)^2 \times m^4$ $9m^6$	n $(2ab)^3 \times (ab)^2$ $8a^5b^5$	o $\frac{(x^3)^2 \times x^4}{x^5}$ x^5	

3 Simplify each of the following.

a	$2(x^3y)^2$	b	$(8t^2u^9v^4)^0$	c	$(-3w^3y)^3$	d	$5(v^4x^2)^3$
	$2x^6y^2$		1		$-27w^9y^3$		$5v^{12}x^6$
e	$-4(p^4qr)^2$	f	$(-5s^7t)^2$	g	$\left(\frac{5w^4y}{2x^3}\right)^2$	h	$\left(\frac{-3x}{2y^3g^5}\right)^2$
	$-4p^8q^2r^2$		$25s^{14}t^2$		$\frac{25w^8y^2}{4x^6}$		$\frac{9x^2}{4y^6g^{10}}$
i	$(3q^6e^4)^4$	j	$\left(\frac{3km^3}{4n^7}\right)^3$	k	$10(3me^5)^2$	l	$\left(\frac{-2r}{n}\right)^4$
	$81q^{24}e^{16}$		$\frac{27k^3m^9}{64n^{21}}$		$90m^2e^{10}$		$\frac{16r^4}{n^4}$
m	$-(-2x^4yz^3)^3$	n	$\left(\frac{-3f}{2^3g^5}\right)^2$	o	$-\left(\frac{-5w^4y}{2zx^3}\right)^2$	p	$2(r^2d^2)^8$
	$8x^{12}y^3z^9$		$\frac{9f^2}{64g^{10}}$		$-\frac{25w^8y^2}{4x^6z^2}$		$2r^{16}d^{16}$
q	$a(3b)^2$	r	$9(p^2q^4)^3$	s	$a(3b^2)^3$	t	$-2y(10x^2ny^3)^3$
	$9ab^2$		$9p^6q^{12}$		$27ab^6$		$-2000x^6n^3y^{10}$
u	$-4w(q^5xk^4)^5$	v	$2x(7x^5b^3y)^2$	w	$-3(2a^3b^4)^2a^2$	x	$2(3x^2y^3)^3$
	$-4wq^{25}x^5k^{20}$		$98x^{11}b^6y^2$		$-12a^8b^8$		$54x^6y^9$
y	$(-4b^2c^5d)^3$	z	$a(2a)^3$	aa	$5a^3(-2a^4b)^3$	bb	$-2(-2^3x^4yz^3)^3$
	$-64b^6c^{15}d^3$		$8a^4$		$-40a^{15}b^3$		$1024x^{12}y^3z^9$

4 Simplify each of the following.

a	$\frac{(2m^3n)^3}{m^4}$	b	$\frac{3(2^2c^4d^5)^3}{(2cd^2)^4}$	c	$\frac{-5(3^5m^3n^2)^2}{(-3^3m^2n)^3}$
	$8m^5n^3$		$12c^8d^7$		$15n$

5 Expand then simplify these expressions.

a $\frac{(v^3x)^3}{(v^2x^2)^4} = \frac{v^9x^3}{v^8x^8} = \frac{v}{x^5}$ ExpandSimplify	b $\frac{(a^5b^2)^4}{(a^2b^3)^2} = \frac{a^{\square}b^{\square}}{a^{\square}b^{\square}} = \frac{\square}{\square}$ $a^{16}b^2$	c $\frac{(n^2a^3)^2}{(n^3a^2)^3} = \frac{n^{\square}a^{\square}}{n^{\square}a^{\square}} = \frac{\square}{\square}$ $\frac{1}{n^5}$
d $\frac{(yx^4)^3}{(y^2x^2)^2}$ $\frac{x^8}{y}$	e $\frac{(v^5c^4)^2}{(v^3c^2)^3}$ $\frac{v}{c^2}$	

6 Simplify the following expressions.

a $\frac{(-2x^2)^3}{-4x}$	b $\frac{(3xy^3)^3}{3x^2y^4}$
$2x^5$	$9xy^5$
c $\frac{(-ab)^3 \times (-ab^2)^2}{-a^5b^3}$	d $\frac{(-2a^3b^2)^2 \times 16a^7b}{(2a^2b)^5}$
b^4	$2a^3$
e $\frac{4m^2n \times 3(m^2n)^3}{6m^2n}$	f $\frac{(7y^2z)^2 \times 3yz^2}{7(yz)^2}$
$2m^6n^3$	$21y^3z^2$
g $\frac{2(ab)^2 \times (2a^2b)^3}{4ab^2 \times 4a^7b^3}$	h $\frac{w^8}{18} \div \left(\frac{w}{6}\right)^2$
1	$2w^6$

7 Simplify the following expressions.

a $\left(\frac{x^2}{y}\right)^3 \times \frac{2x}{y^4}$

$$\frac{2x^7}{y^7}$$

b $\left(\frac{a^2b}{c}\right)^4 \times \left(\frac{b^2}{c}\right)^3$

$$\frac{a^8b^{10}}{c^7}$$

c $\left(\frac{pq^3}{r^2}\right)^2 \times \left(\frac{p^0q^2}{r}\right)^4$

$$\frac{p^2q^{14}}{r^8}$$

d $\left(\frac{a^3b}{c}\right)^3 \times \left(\frac{ac^4}{b}\right)^2$

$$a^{11}bc^5$$

e $\left(\frac{x^2z}{y}\right)^4 \times \left(\frac{xy^2}{z}\right)^3$

$$x^{11}y^2z$$

f $\left(\frac{r^3s}{t}\right)^2 \div \left(\frac{s}{rt^4}\right)^3$

$$\frac{r^9t^{10}}{s}$$

8 Simplify the following expressions.

a $\frac{3a \times 3a \times 3a}{3a + 3a + 3a}$

$$3a^2$$

b $\frac{3c \times 4c^2 \times 5c^3}{3c + 4c^2 + 5c^3}$

$$\frac{60c^4}{3 + 4c + 5c^2}$$

c $\frac{ab^2 \times 2b^2c^3 \times 3c^3a^4}{a^3b^3 + 2a^3b^3 + 3a^3b^3}$

$$a^2bc^6$$